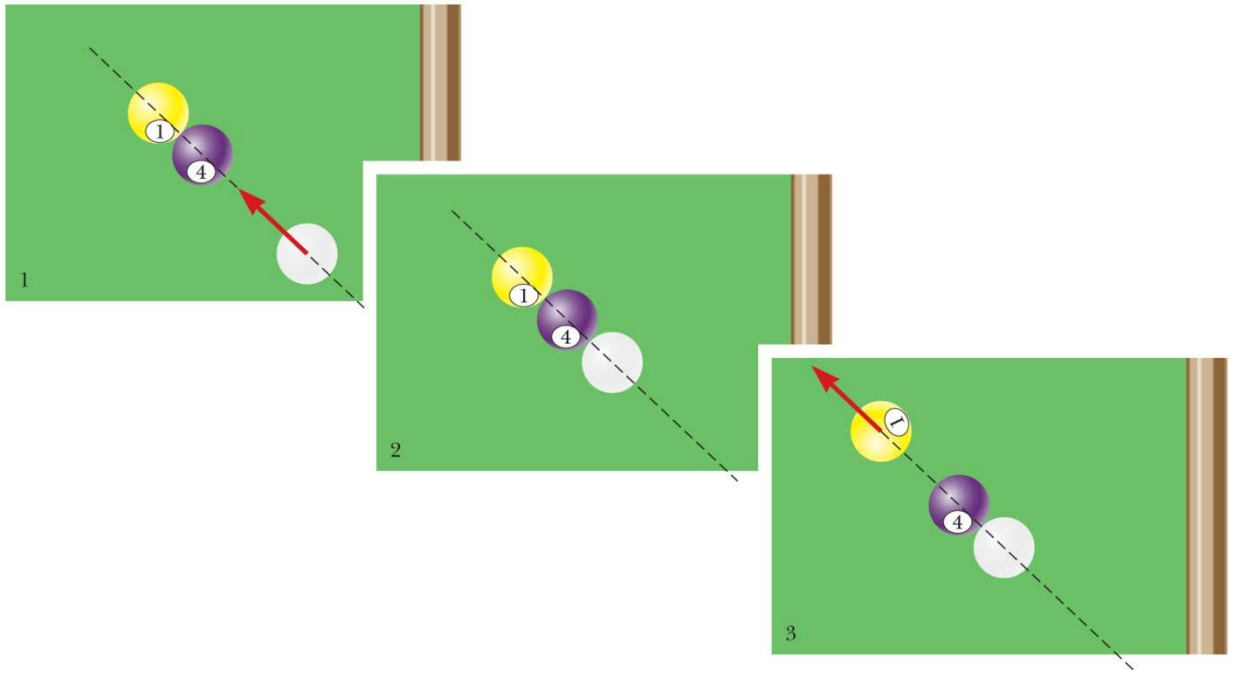
Storyline

Chapter 9:
Linear Momentum and Collisions



Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

BROOKS/COLE
CENGAGE Learning

Reading Question 10.1

Which of these is *not* a vector? Pick the scalar quantity from the list.

a. velocity
b. force
c. momentum
d. energy

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Reading Question 10.1

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Reading Question 10.2

A prototype consumer drone aircraft has a momentum of 230 kg m/s as it passes over a monitoring station. Which of these modifications would give it four times the original momentum?

a. Double the velocity
b. Quadruple the velocity (four times larger)
c. Halve the mass
d. Quarter the mass

(assuming everything else is constant)

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Reading Question 10.3

A 1.0 kg steel sphere is at the origin, and a 0.1 kg sphere of brass is on the *x* axis at 0.1 m. The center of mass of this two particle system is…

a. undefined if the relative velocity is unknown

b. on the *x* axis, closer to the steel sphere at the origin

c. on the *x* axis, closer to the brass sphere at 0.1 m

d. at the center of the heavier steel sphere

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Reading Question 10.3

A 1.0 kg steel sphere is at the origin, and a 0.1 kg sphere of brass is on the *x* axis at 0.1 m.

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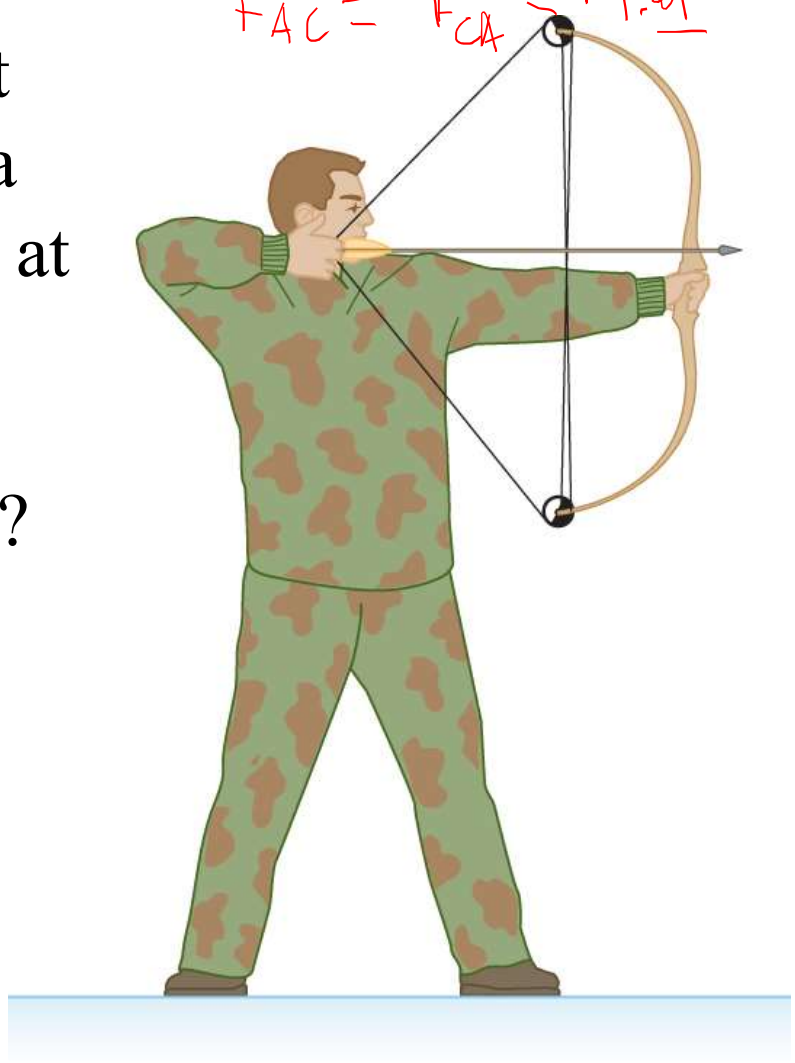
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Linear Momentum

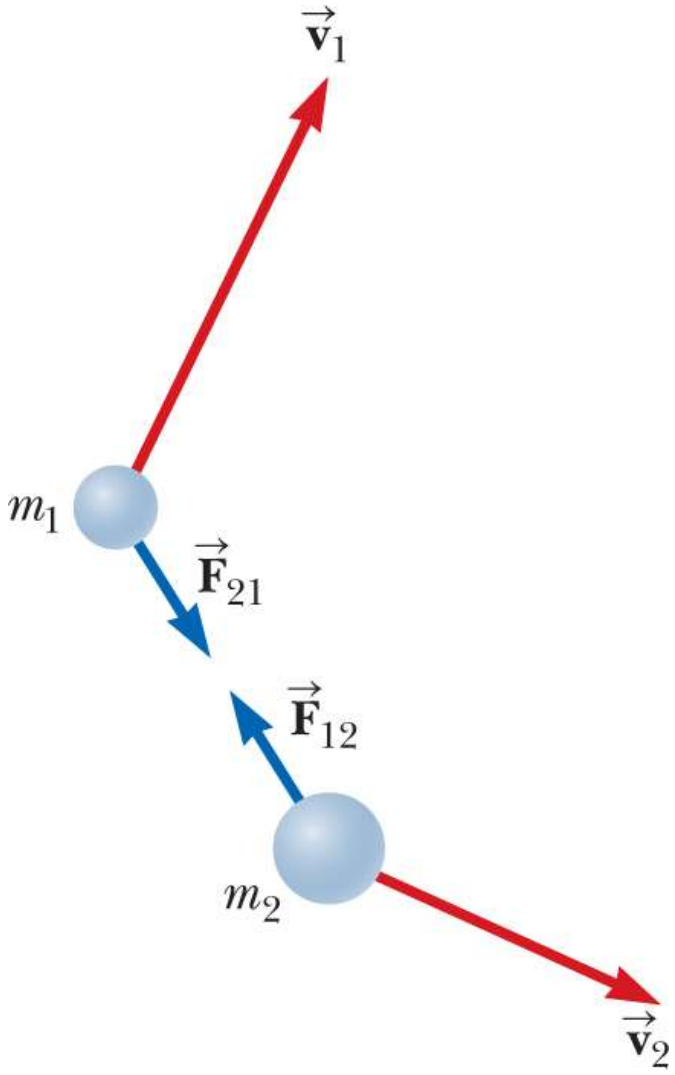
$v^2 = 2ax$

A 60-kg archer stands at rest on frictionless ice and fires a 0.030-kg arrow horizontally at 85 m/s. With what velocity does the archer move across the ice after firing the arrow?

$F_{AC} = -F_{CA} = m \cdot a$



Linear Momentum

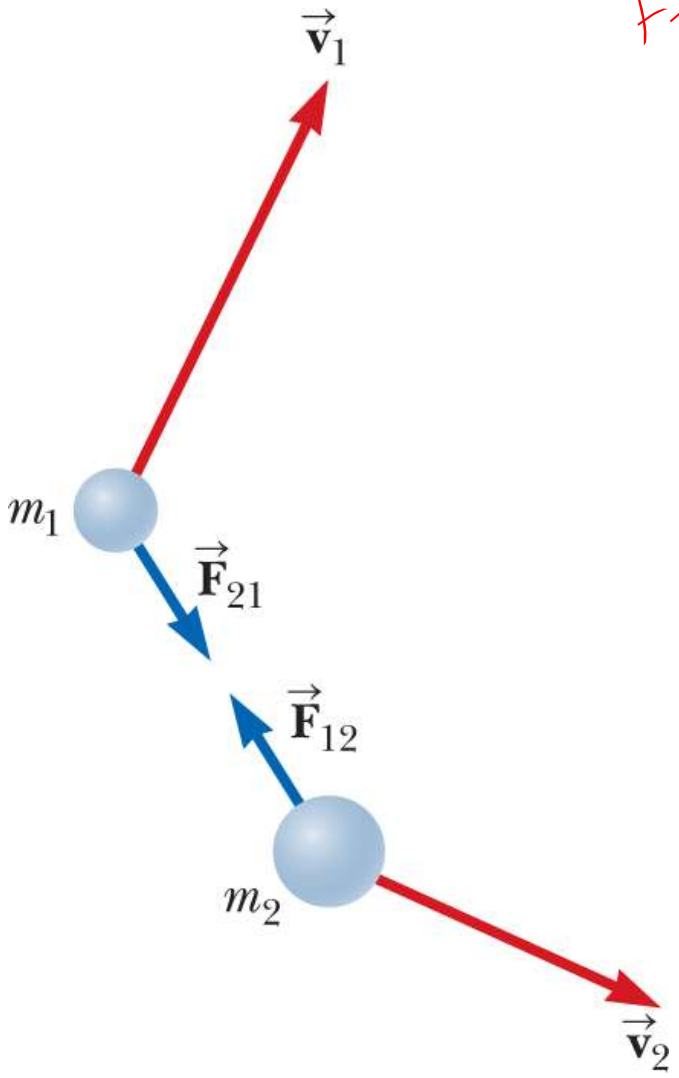


System is isolated:
internal forces only

$$\vec{\mathbf{F}}_{12} = -\vec{\mathbf{F}}_{21}$$

$$\vec{\mathbf{F}}_{12} + \vec{\mathbf{F}}_{21} = 0 = \sum \vec{F}_{\text{int ext}}$$

Linear Momentum



$$m_1\vec{\mathbf{a}}_1 + m_2\vec{\mathbf{a}}_2 = 0$$

$$m_1\frac{d\vec{\mathbf{v}}_1}{dt} + m_2\frac{d\vec{\mathbf{v}}_2}{dt} = 0$$

$$\frac{d\left(m_1\vec{\mathbf{v}}_1\right)}{dt} + \frac{d\left(m_2\vec{\mathbf{v}}_2\right)}{dt} = 0$$

$$\frac{d}{dt}\left(m_1\vec{\mathbf{v}}_1 + m_2\vec{\mathbf{v}}_2\right) = 0$$

Linear Momentum

The **linear momentum** $\vec{\mathbf{p}}$ of a particle or an object that can be modeled as a particle of mass m moving with a velocity $\vec{\mathbf{v}}$ is defined to be the product of the mass and velocity of the particle:

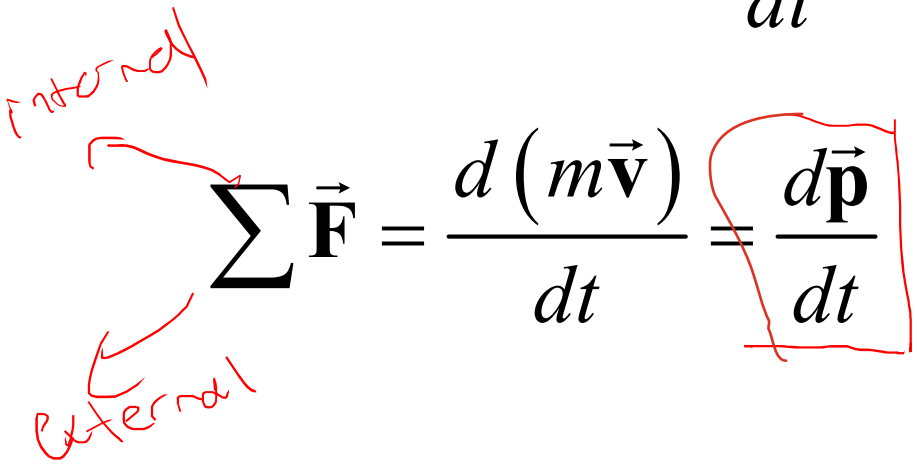
$$\vec{\mathbf{p}} \equiv m\vec{\mathbf{v}}$$

$$p_x = mv_x \quad p_y = mv_y \quad p_z = mv_z$$

Linear Momentum and Force

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}} = m\frac{d\vec{\mathbf{v}}}{dt}$$

$$\sum \vec{\mathbf{F}} = \frac{d\left(m\vec{\mathbf{v}}\right)}{dt} = \frac{d\vec{\mathbf{p}}}{dt}$$



The time rate of change of the linear momentum of a particle is equal to the net force acting on the particle

Quick Quiz 9.1

Two objects have equal kinetic energies. How do the magnitudes of their momenta compare?

(a) $p_1 < p_2$

(b) $p_1 = p_2$

(c) $p_1 > p_2$

(d) not enough information to tell

Quick Quiz 9.1

Two objects have equal kinetic energies. How do the magnitudes of their momenta compare?

$m_1 = m_2$ / $v_1 = v_2$

$K_1 = K_2$

$p_1 = ?_2$

(a) $p_1 < p_2$

(b) $p_1 = p_2$

(c) $p_1 > p_2$

(d) **not enough information to tell**

Quick Quiz 9.2

Your physical education teacher throws a baseball to you at a certain speed and you catch it. The teacher is next going to throw you a medicine ball whose mass is ten times the mass of the baseball. You are given the following choices: You can have the medicine ball thrown with

(a) the same speed

(b) the same momentum

(c) the same kinetic energy

as the baseball.

Rank these choices from easiest to hardest to catch.

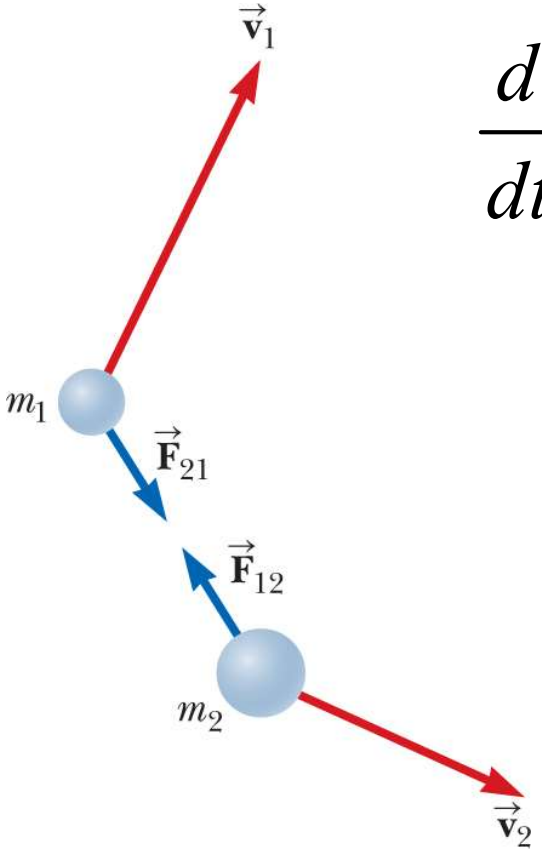
Quick Quiz 9.2

(1) the same momentum

(2) the same kinetic energy

(3) the same speed

Analysis Model: Isolated System (Momentum)



$$\frac{d}{dt}\left(m_1\vec{\mathbf{v}}_1 + m_2\vec{\mathbf{v}}_2\right) = 0 \Rightarrow \frac{d}{dt}\left(\vec{\mathbf{p}}_1 + \vec{\mathbf{p}}_2\right) = 0$$

$$\vec{\mathbf{p}}_{\text{tot}} = \text{constant}$$

$$\vec{\mathbf{p}}_{\text{tot}} = 0$$

$$\vec{\mathbf{p}}_{1i} + \vec{\mathbf{p}}_{2i} = \vec{\mathbf{p}}_{1f} + \vec{\mathbf{p}}_{2f}$$

$$p_{1ix} + p_{2ix} = p_{1fx} + p_{2fx} \quad p_{1iy} + p_{2iy} = p_{1fy} + p_{2fy}$$

$$p_{1iz} + p_{2iz} = p_{1fz} + p_{2fz}$$

Analysis Model: Isolated System (Momentum)

Whenever two or more particles in an isolated system interact, the total momentum of the system does not change.



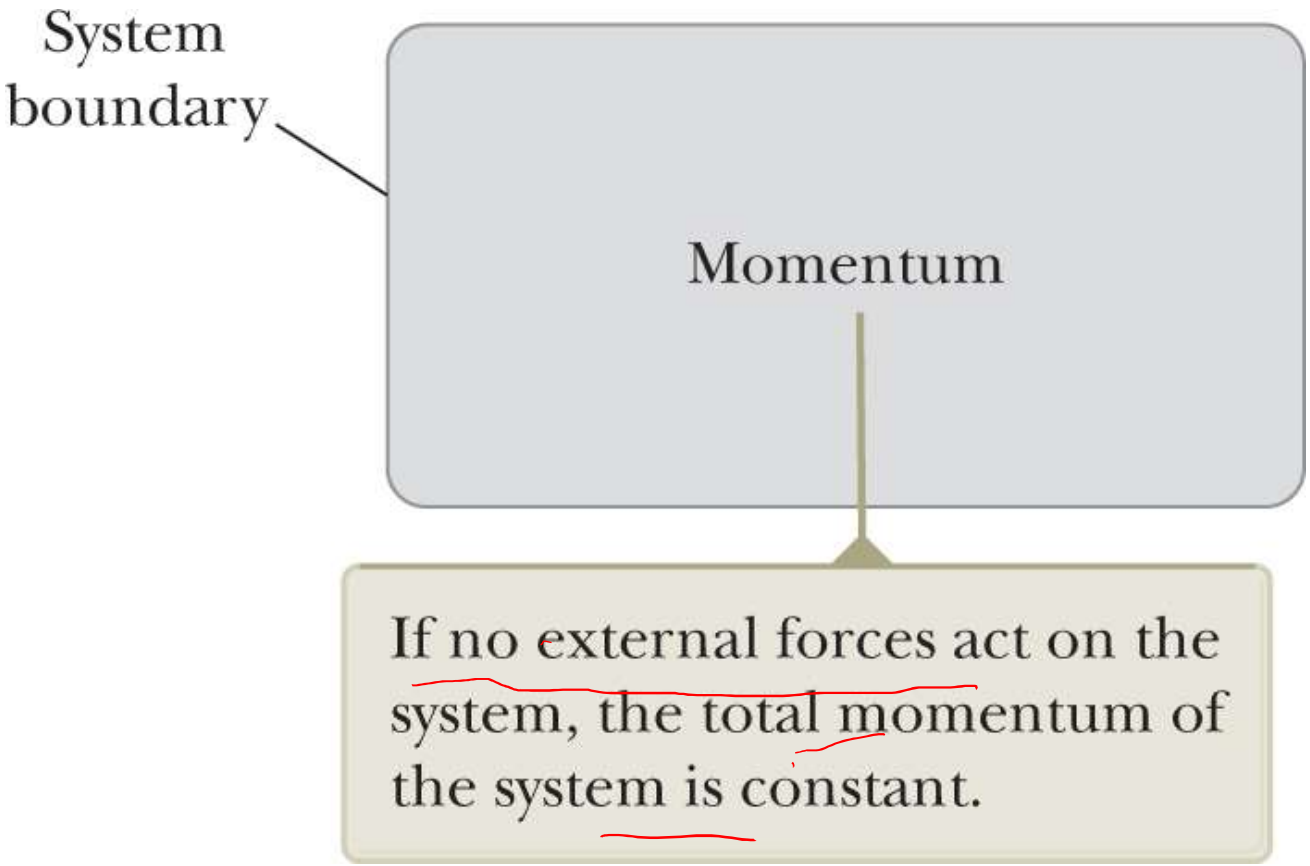
For *energy*: system isolated if there are no transfers of energy across boundary of system

For *momentum*: must be no external forces on system

If there are external forces ptotal changes.

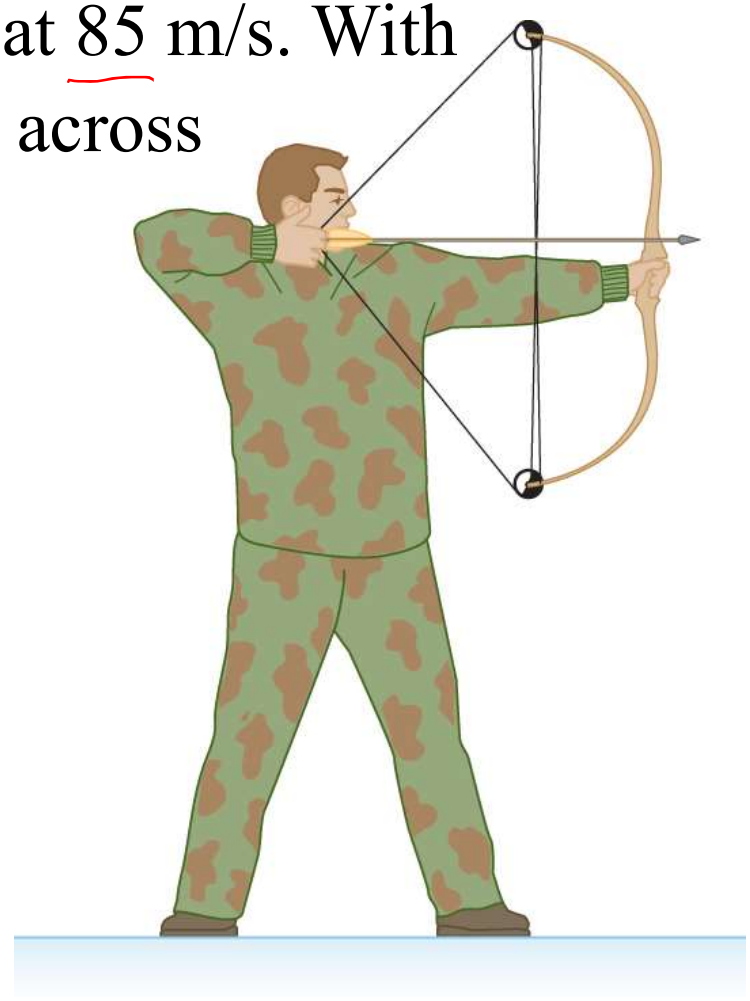
Analysis Model: Isolated System (Momentum)

$$\Delta \vec{\mathbf{p}}_{\text{tot}} = 0$$



Example 9.1:
The Archer

A 60-kg archer stands at rest on frictionless ice and fires a 0.030-kg arrow horizontally at 85 m/s. With what velocity does the archer move across the ice after firing the arrow?



Example 9.1: The Archer

$m_1 = 60$ kg, $m_2 = 0.030$ kg, and $\vec{\mathbf{v}}_{2f} = 85\hat{\mathbf{i}}$ m/s

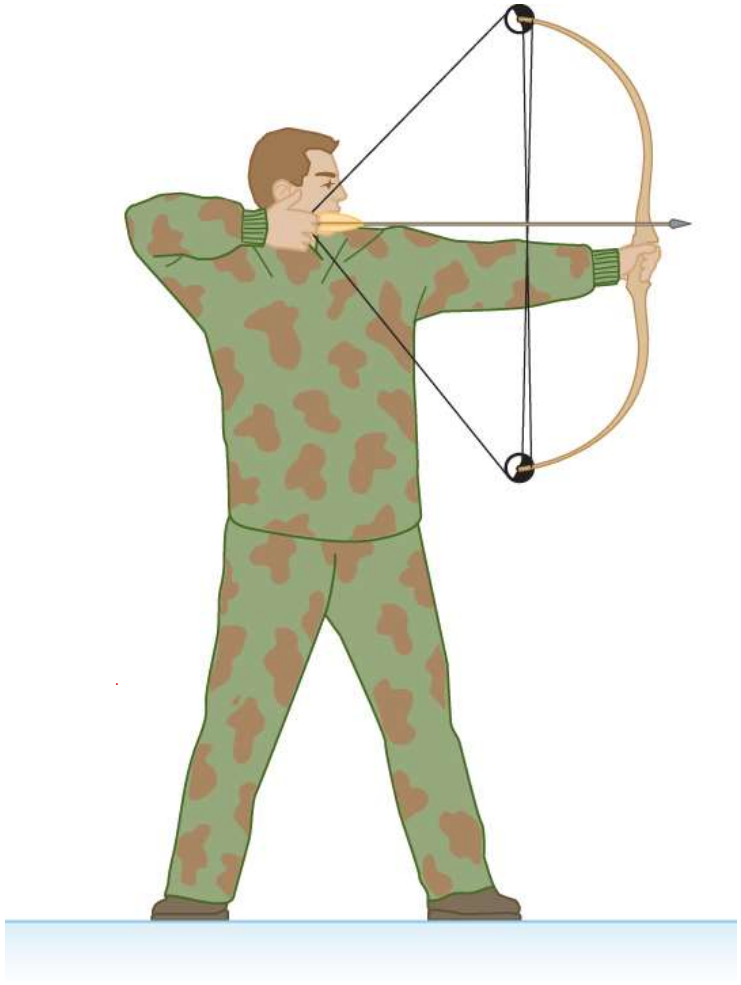
$$\Delta\vec{\mathbf{p}} = 0 \rightarrow \vec{\mathbf{p}}_f - \vec{\mathbf{p}}_i = 0 \Rightarrow \vec{\mathbf{p}}_f = \vec{\mathbf{p}}_i$$

$$\Rightarrow m_1\vec{\mathbf{v}}_{1f} + m_2\vec{\mathbf{v}}_{2f} = 0$$

$$\vec{\mathbf{v}}_{1f} = -\frac{m_2}{m_1}\vec{\mathbf{v}}_{2f}$$

$$= -\left(\frac{0.030 \text{ kg}}{60 \text{ kg}}\right)\left(85\ \hat{\mathbf{i}} \text{ m/s}\right)$$

$$= -4.25\ \hat{\mathbf{i}} \text{ cm/s}$$



Example 9.1:
The Archer

What if the arrow were fired in a direction that makes an angle θ with the horizontal? How will that change the recoil velocity of the archer?



$$m_1 v_{1f} + m_2 v_{2f} \cos\theta = 0$$

$$v_{1f} = -\frac{m_2}{m_1} v_{2f} \cos\theta = -\left(4.25 \text{ cm/s } \hat{\mathbf{i}}\right)\cos\theta$$

Example 9.2: Can We Really Ignore the Kinetic energy of the Earth?

Earlier, we claimed that we can ignore the kinetic energy of the Earth when considering the energy of a system consisting of the Earth and a dropped ball. Verify this claim.

Example 9.2: Can We Really Ignore the Kinetic Energy of the Earth

$$\frac{K_E}{K_b} = \frac{\frac{1}{2} m_E v_E^{\,2}}{\frac{1}{2} m_b v_b^{\,2}} = \left(\frac{m_E}{m_b}\right)\left(\frac{v_E}{v_b}\right)^2$$

$$\Delta\vec{\mathbf{p}} = 0 \rightarrow p_i = p_f$$

$$\Rightarrow 0 = m_b v_b + m_E v_E$$

$$\frac{v_E}{v_b} = -\frac{m_b}{m_E}$$

$$\frac{K_E}{K_b} = \left(\frac{m_E}{m_b}\right)\left(\frac{m_b}{m_E}\right)^2 = \frac{m_b}{m_E}$$

$$\frac{K_E}{K_b} = \frac{m_b}{m_E} \sim \frac{1\ \text{kg}}{10^{25}\ \text{kg}} \sim 10^{-25}$$

Analysis Model: Nonisolated System (Momentum)

$$\sum \vec{\mathbf{F}}_{ext} = \frac{d\vec{\mathbf{p}}}{dt}$$

$\sum F_{external} = 0$

$\frac{d\vec{p}}{dt} = 0$

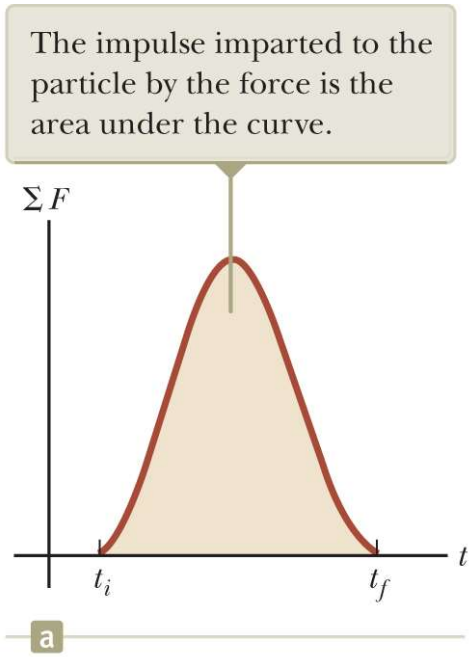
$p_{initial} = p_{final}$

$$d\vec{\mathbf{p}} = \sum \vec{\mathbf{F}}_{ext} dt$$

$$\Delta\vec{\mathbf{p}} = \left|\vec{\mathbf{p}}_f - \vec{\mathbf{p}}_i\right| = \int_{t_i}^{t_f} \sum \vec{\mathbf{F}}_{ext} dt$$

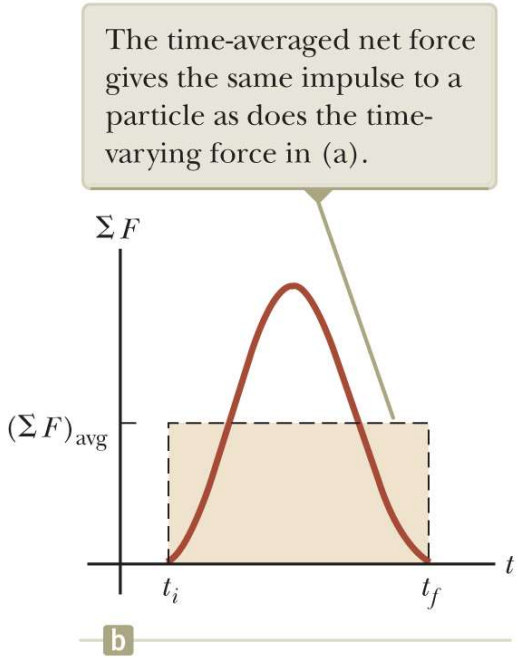
Impulse: $\vec{\mathbf{I}} \equiv \int_{t_i}^{t_f} \sum \vec{\mathbf{F}} dt$

Impulse



$$\left(\sum \vec{\mathbf{F}}\right)_{\text{avg}} \equiv \frac{1}{\Delta t}\int_{t_i}^{t_f} \sum \vec{\mathbf{F}}\, dt$$

$$\vec{\mathbf{I}} = \left(\sum \vec{\mathbf{F}}\right)_{\text{avg}} \Delta t$$



$$\left(\sum \vec{\mathbf{F}}\right)_{\text{avg}} = \sum \vec{\mathbf{F}} \quad \left(\vec{\mathbf{F}} \text{ constant}\right)$$

$$\vec{\mathbf{I}} = \sum \vec{\mathbf{F}}\, \Delta t \quad \left(\text{constant net force}\right)$$

Impulse and Momentum

The change in the momentum of a particle is equal to the impulse of the net force acting on the particle:

$$\Delta\vec{\mathbf{p}} = \vec{\mathbf{I}}$$

Impulse, Momentum, and Average Force



Impulse approximation: assume one of the forces exerted on particle acts for a short time but is much greater than any other force present

$$\Delta\vec{\mathbf{p}} = \vec{\mathbf{I}} = \left(\sum \vec{\mathbf{F}}\right)_{\text{avg}} \Delta t$$

Quick Quiz 9.3 Part I

Two objects are at rest on a frictionless surface.

Object 1 has a greater mass than object 2. $m_1 > m_2$

When a constant force is applied to object 1, it accelerates through a distance d in a straight line. F

The force is removed from object 1 and is applied to object 2. At the moment when object 2 has accelerated through the same distance d, which statements are true?

(a) $p_1 < p_2$ (b) $p_1 = p_2$ (c) $p_1 > p_2$

(d) $K_1 < K_2$ (e) $K_1 = K_2$ (f) $K_1 > K_2$

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(d) $K_1 < K_2$ **(e) $K_1 = K_2$** (f) $K_1 > K_2$

Quick Quiz 9.3 Part II

Two objects are at rest on a frictionless surface.

Object 1 has a greater mass than object 2. $m_1 > m_2$

When a force is applied to object 1, it accelerates for a time interval Δt.

The force is removed from object 1 and is applied to object 2 for the same time interval Δt. Which statements are true?

(a) $p_1 < p_2$ (b) $p_1 = p_2$ (c) $p_1 > p_2$

(d) $K_1 < K_2$ (e) $K_1 = K_2$ (f) $K_1 > K_2$

Quick Quiz 9.3 Part II

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Quick Quiz 9.4 Part I

An automobile dashboard, seat belt, and air bag, are each used alone in separate collisions from the same speed. In terms of impulse, rank these from greatest to least.

Quick Quiz 9.4 Part I

An automobile dashboard, seat belt, and air bag, are each used alone in separate collisions from the same speed. In terms of impulse, rank these from greatest to least.

All three are the same.

Quick Quiz 9.4 Part II

An automobile dashboard, seat belt, and air bag, are each used alone in separate collisions from the same speed. In terms of the average force each delivers to a front-seat passenger, rank these from greatest to least.

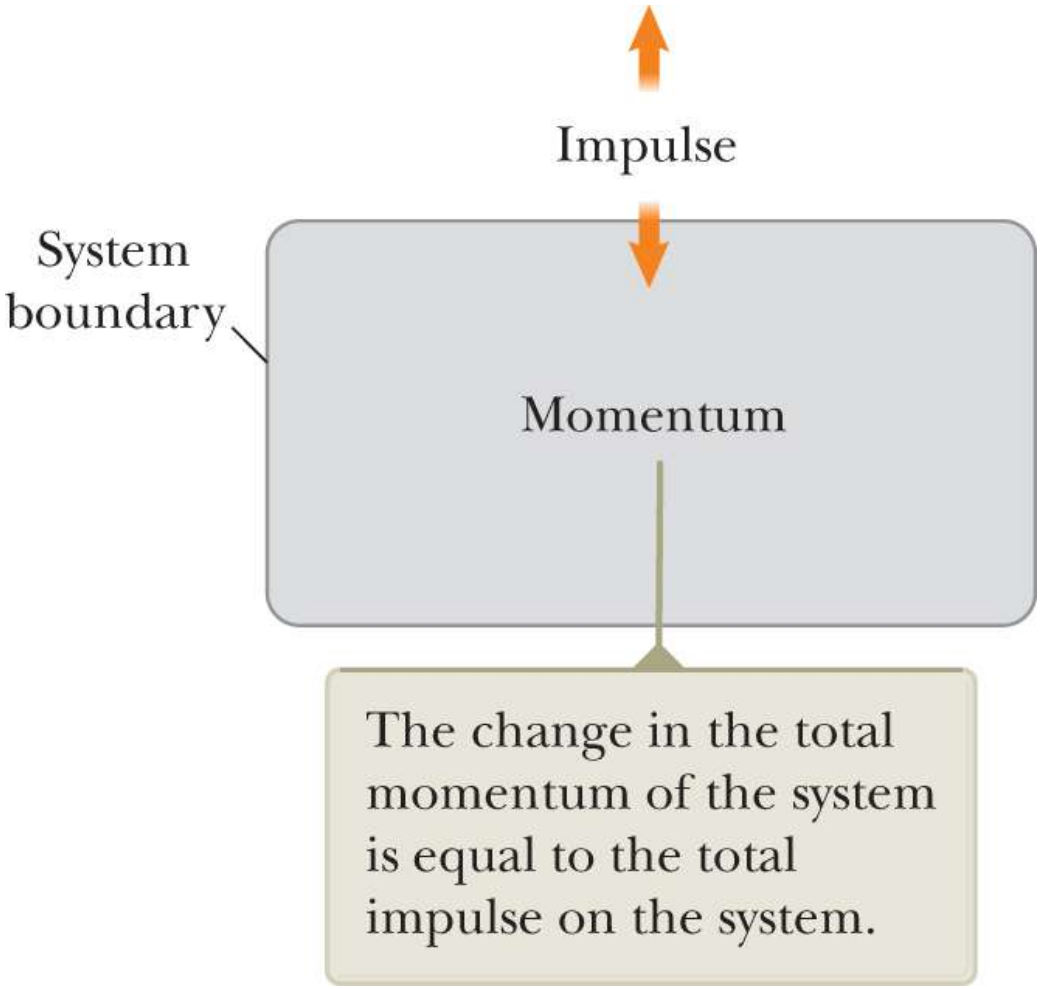
Quick Quiz 9.4 Part II

An automobile dashboard, seat belt, and air bag, are each used alone in separate collisions from the same speed. In terms of the average force each delivers to a front-seat passenger, rank these from greatest to least.

dashboard, seat belt, air bag

Analysis Model: Nonisolated System (Momentum)

$$\Delta\vec{\mathbf{p}} = \vec{\mathbf{I}}$$

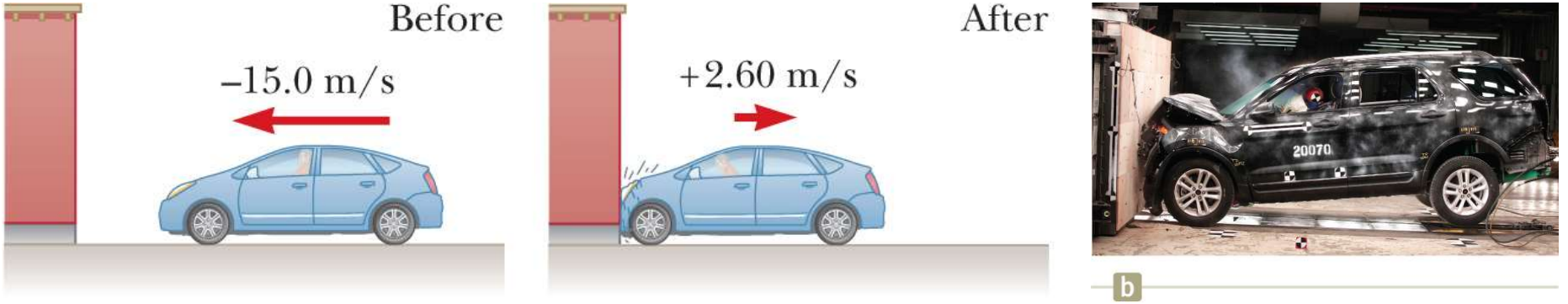


Example 9.3:
How Good Are the Bumpers?

In a particular crash test, a car of mass 1 500 kg collides with a wall as shown in the figure. The initial and final velocities of the car are

$$\vec{\mathbf{v}}_i = -15.0\ \mathbf{i}\ \text{m/s and } \vec{\mathbf{v}}_f = 2.60\ \mathbf{i}\ \text{m/s},$$

respectively. If the collision lasts 0.150 s, find the impulse on the car during the collision and the average net force exerted on the car.



Example 9.3:
How Good Are the Bumpers?

$$\vec{\mathbf{I}} = \Delta\vec{\mathbf{p}} = \Delta\vec{\mathbf{p}}_f - \Delta\vec{\mathbf{p}}_i = m\vec{\mathbf{v}}_f - m\vec{\mathbf{v}}_i = m\left(\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i\right)$$

$$= (1500\ \text{kg})\left[2.60\ \hat{\mathbf{i}}\ \text{m/s} - \left(-15.0\ \hat{\mathbf{i}}\ \text{m/s}\right)\right]$$

$$= \boxed{2.64\times10^4\ \hat{\mathbf{i}}\ \text{kg}\cdot\text{m/s}}$$

$$\left(\sum\vec{\mathbf{F}}\right)_{\text{avg}} = \frac{\vec{\mathbf{I}}}{\Delta t} = \frac{2.64\times10^4\ \hat{\mathbf{i}}\ \text{kg}\cdot\text{m/s}}{0.150\ \text{s}} = 1.76\times10^5\ \hat{\mathbf{i}}\ \text{N}$$

Example 9.3:
How Good Are the Bumpers?

What if the car did not rebound from the wall?
Suppose the final velocity of the car is zero and the time interval of the collision remains at 0.150 s. Would that represent a larger or a smaller net force on the car?

$$\begin{aligned}\vec{\mathbf{I}} &= \Delta\vec{\mathbf{p}} = \Delta\vec{\mathbf{p}}_f - \Delta\vec{\mathbf{p}}_i \\ &= 0 - (1500\ \text{kg})(-15.0\ \hat{\mathbf{i}}\ \text{m/s}) \\ &= 2.25\times10^4\ \hat{\mathbf{i}}\ \text{kg}\cdot\text{m/s}\end{aligned}$$

$$\left(\sum\vec{\mathbf{F}}\right)_{\text{avg}} = \frac{\vec{\mathbf{I}}}{\Delta t} = \frac{2.25\times10^4\ \hat{\mathbf{i}}\ \text{kg}\cdot\text{m/s}}{0.150\ \text{s}} = 1.50\times10^5\ \hat{\mathbf{i}}\ \text{N}$$

smaller net force

Reading Question 11.1

During a collision, many external forces acting on a system can be neglected because they're weaker than the interactions involved in the collision.

What is this called?

a. An elastic collision
b. An inelastic collision
c. Impulse conservation
d. Impulse approximation

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Reading Question 11.1

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What is this called?

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Reading Question 11.2

Which of these descriptions best captures the idea of impulse?

a. Change in velocity of a particle
b. Change in velocity of a system's center of mass
c. Integral of force through time
d. Integral of momentum through time

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Reading Question 11.2

Which of these descriptions best captures the idea of impulse?

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c. Integral of force through time
d. Integral of momentum through time

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Reading Question 11.3

Momentum can be conserved in two-dimensional collisions, but kinetic energy is only conserved in some one dimensional collisions.

a. True

b. False because kinetic energy can be conserved in 2-d collisions

c. False because momentum is only conserved when all three dimensions are included

d. False because kinetic energy can only be conserved if all three dimensions are included

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Reading Question 11.3

Momentum can be conserved in two-dimensional collisions, but kinetic energy is only conserved in some one dimensional collisions.

a. True

b. False because kinetic energy can be conserved in 2-d collisions

c. False because momentum is only conserved when all three dimensions are included

d. False because kinetic energy can only be conserved if all three dimensions are included

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Reading Question 11.4

A completely inelastic collision features...

a. Particles that stick together
b. Particles that break and move apart rapidly
c. A target particle that begins at rest
d. Particles that leave the collision in directions 90° apart.

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Reading Question 11.4

A completely inelastic collision features...

a. Particles that stick together

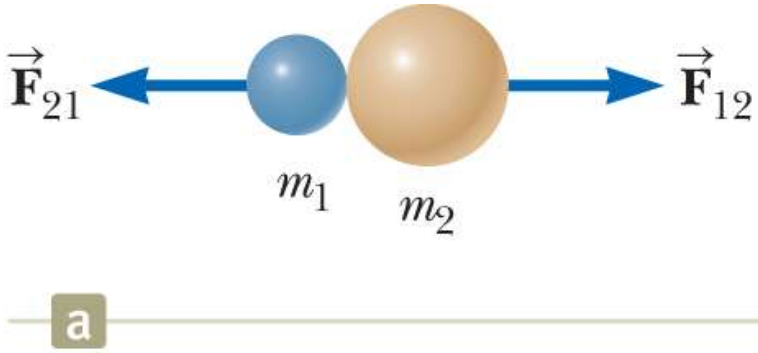
b. Particles that break and move apart rapidly

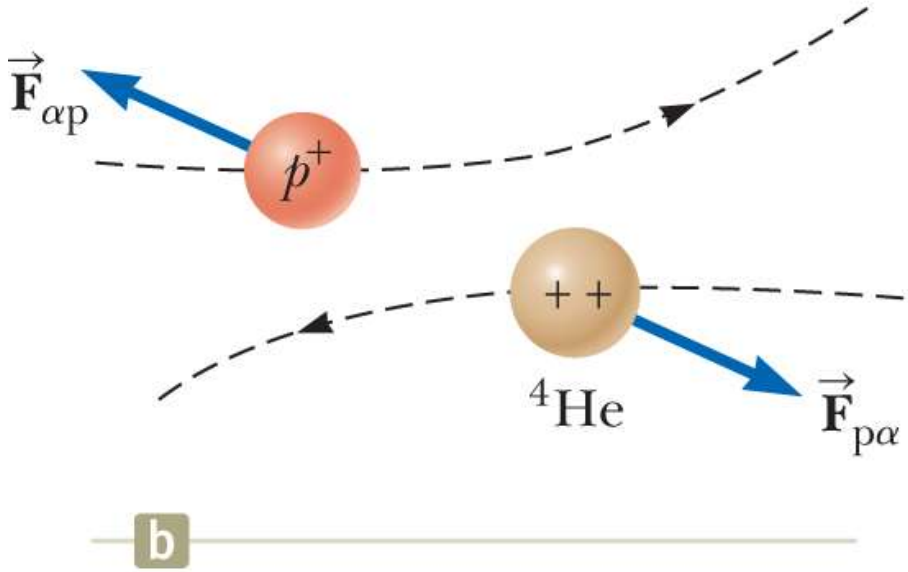
c. A target particle that begins at rest

d. Particles that leave the collision in directions 90° apart.

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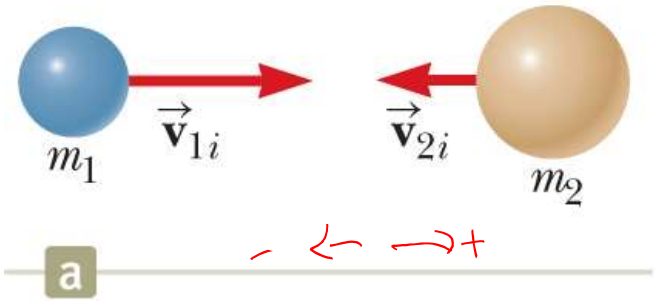
Collisions in One Dimension



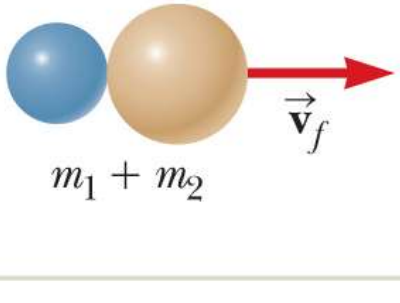


Perfectly Inelastic Collisions

Before the collision, the particles move separately.



After the collision, the particles move together.



$$\Delta\vec{\mathbf{p}} = 0$$

$$\Delta\vec{\mathbf{p}}_i = \Delta\vec{\mathbf{p}}_f$$

$$m_1\vec{\mathbf{v}}_{1i} + m_2\vec{\mathbf{v}}_{2i} = \left(m_1 + m_2\right)\vec{\mathbf{v}}_f$$

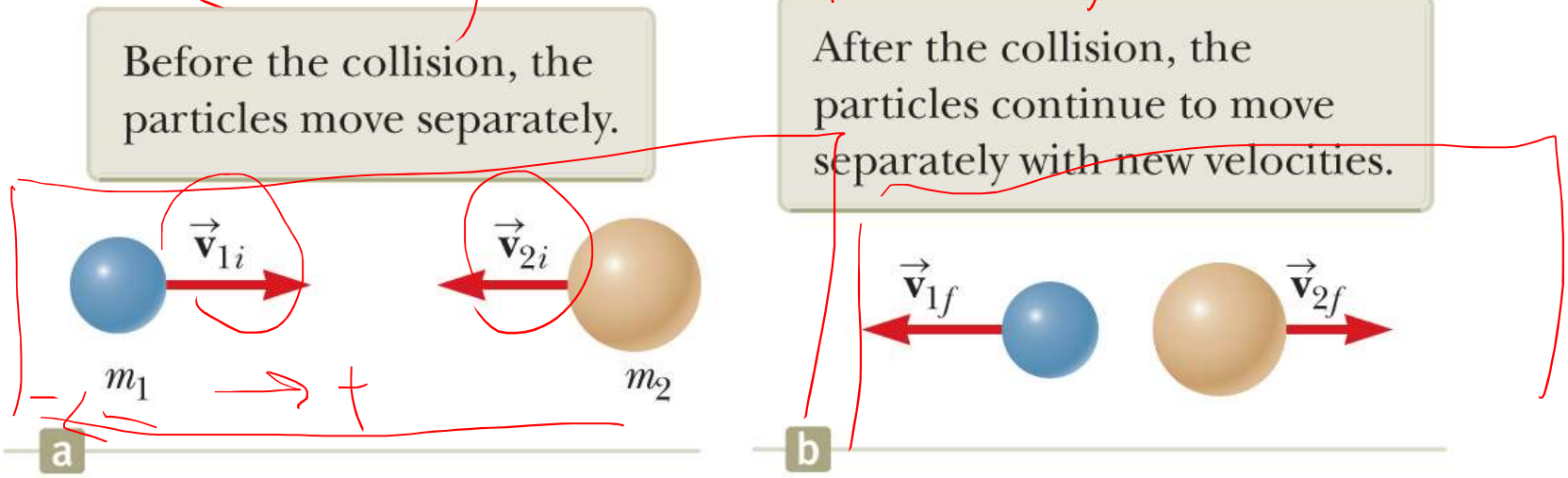
$$\vec{\mathbf{v}}_f = \frac{m_1\vec{\mathbf{v}}_{1i} + m_2\vec{\mathbf{v}}_{2i}}{m_1 + m_2}$$

Elastic Collisions

$$p_i = p_f \rightarrow m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$K_i = K_f \rightarrow \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

$$(v_{1i} - v_{2i}) = -(v_{1f} - v_{2f})$$



Elastic Collisions

$$\frac{1}{2}m_1 v_{1i}^{\ 2} + \frac{1}{2}m_2 v_{2i}^{\ 2} = \frac{1}{2}m_1 v_{1f}^{\ 2} + \frac{1}{2}m_2 v_{2f}^{\ 2}$$

$$m_1\left(v_{1i}^{\ 2} - v_{1f}^{\ 2}\right) = m_2\left(v_{2f}^{\ 2} - v_{2i}^{\ 2}\right)$$

$$m_1\left(v_{1i} - v_{1f}\right)\left(v_{1i} + v_{1f}\right) = m_2\left(v_{2f} - v_{2i}\right)\left(v_{2f} + v_{2i}\right)$$

$$p_i = p_f \rightarrow m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$m_1\left(v_{1i} - v_{1f}\right) = m_2\left(v_{2f} - v_{2i}\right)$$

güzel frmül

$$v_{1i} + v_{1f} = v_{2f} + v_{2i} \quad \Rightarrow \quad v_{1i} - v_{2i} = -\left(v_{1f} - v_{2f}\right)$$

3

Elastic Collisions

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$v_{1i} - v_{2i} = -\left(v_{1f} - v_{2f}\right)$$

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2}\right) v_{2i}$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) v_{2i}$$

Elastic Collisions

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2}\right) v_{2i}$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) v_{2i}$$

$m_1 = m_2 = m$

$$v_{1f} = \left(\frac{m - m}{m + m}\right) v_{1i} + \left(\frac{2m}{m + m}\right) v_{2i} = v_{2i}$$

$$v_{2f} = \left(\frac{2m}{m + m}\right) v_{1i} + \left(\frac{m - m}{m + m}\right) v_{2i} = v_{1i}$$

Elastic Collisions

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2}\right) v_{2i}$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) v_{2i}$$

If particle 2 is at rest, $v_{2i} = 0$:

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{1i}$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i}$$

$$m_1 >> m_2 \rightarrow m_1 + m_2 \approx m_1 \text{ and } m_1 - m_2 \approx m_1$$

$$v_{1f} \approx \left(\frac{m_1}{m_1}\right) v_{1i} \text{ and } v_{2f} \approx \left(\frac{2m_1}{m_1}\right) v_{1i}$$

$$\Rightarrow v_{1f} \approx v_{1i} \text{ and } v_{2f} \approx 2v_{1i}$$

Elastic Collisions

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} \text{ and } v_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) v_{1i}$$

$$m_1 << m_2 \rightarrow m_1 + m_2 \approx m_2 \text{ and } m_1 - m_2 \approx -m_2$$

$$v_{1f} \approx \left(\frac{-m_2}{m_2} \right) v_{1i} \text{ and } v_{2f} \approx 0$$

$$\Rightarrow v_{1f} \approx -v_{1i} \text{ and } v_{2f} \approx 0$$

Quick Quiz 9.5

In a perfectly inelastic one-dimensional collision between two moving objects, what condition alone is necessary so that the final kinetic energy of the system is zero after the collision?

(a) The objects must have initial momenta with the same magnitude but opposite directions.

(b) The objects must have the same mass.

(c) The objects must have the same initial velocity.

(d) The objects must have the same initial speed, with velocity vectors in opposite directions.

Quick Quiz 9.5

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Quick Quiz 9.6

A table-tennis ball is thrown at a stationary bowling ball. The table-tennis ball makes a one-dimensional elastic collision and bounces back along the same line. Compared with the bowling ball after the collision, the table-tennis ball has

(a) a larger magnitude of momentum and more kinetic energy.

(b) a smaller magnitude of momentum and more kinetic energy.

(c) a larger magnitude of momentum and less kinetic energy.

(d) a smaller magnitude of momentum and less kinetic energy.

(e) the same magnitude of momentum and the same kinetic energy.

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(d) a smaller magnitude of momentum and less kinetic energy.

(e) the same magnitude of momentum and the same kinetic energy.

Problem-Solving Strategy: One-Dimensional Collisions

1. Conceptualize
2. Categorize
3. Analyze

perfectly inelastic collision:

$$\vec{\mathbf{v}}_f = \frac{m_1\vec{\mathbf{v}}_{1i} + m_2\vec{\mathbf{v}}_{2i}}{m_1 + m_2}$$

Beraber gidiyorlar.

kinetik enerji korunmaz.

elastic collision:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \text{ and } v_{1i} - v_{2i} = -\left(v_{1f} - v_{2f}\right)$$

ikisi de sekiyor

Kinetik enerji korunur.

inelastic collision:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

kinetic energy is not conserved

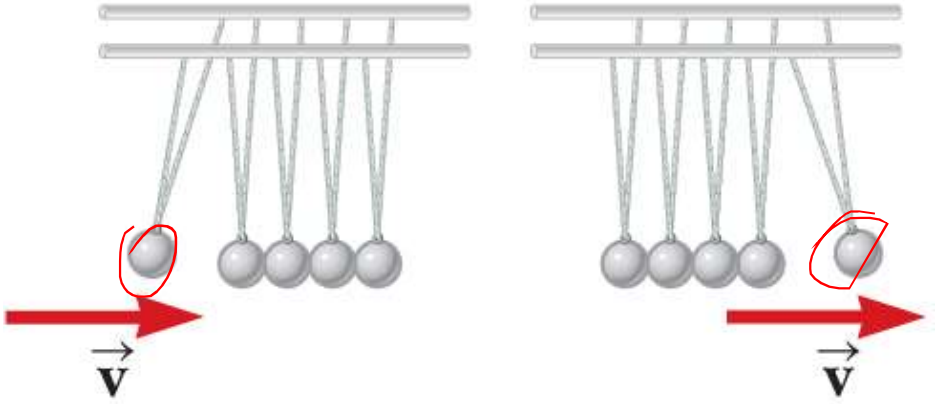
4. Finalize

sekerler ama arada ısıya, sese giden enerji kaybı olur.

Example 9.4:
The Executive Stress Reliever

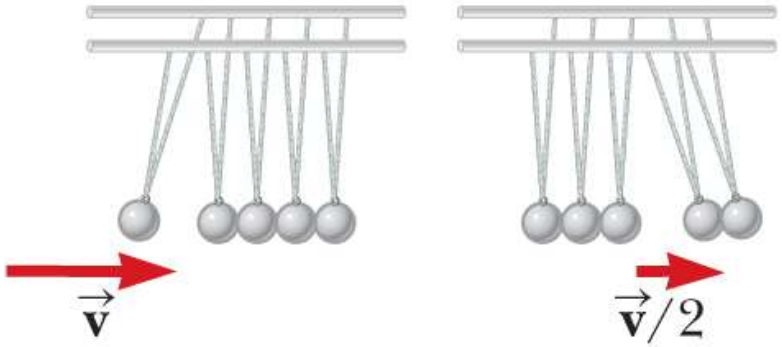


An ingenious device that illustrates conservation of momentum and kinetic energy is shown in the photo. It consists of five identical hard balls supported by strings of equal lengths. When ball 1 is pulled out and released, after the almost-elastic collision between it and ball 2, ball 1 stops and ball 5 moves out as shown in the lower figure. If balls 1 and 2 are pulled out and released, they stop after the collision and balls 4 and 5 swing out, and so forth. Even if four balls (1–4) are pulled out and released, four balls (2 – 5) swing out after the collision!



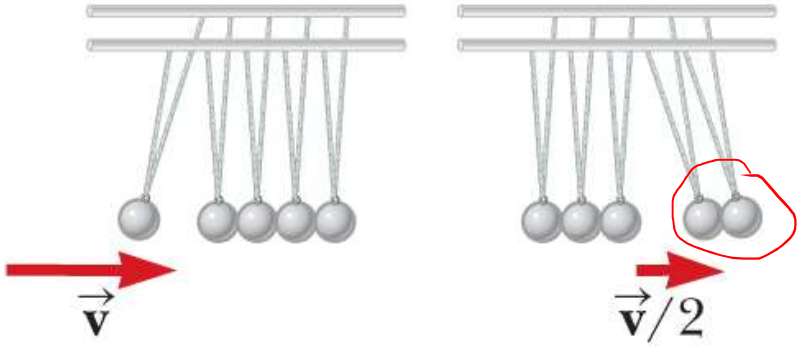
Example 9.4:
The Executive Stress Reliever

Is it ever possible that when ball 1 is released, it stops after the collision and balls 4 and 5 will swing out on the opposite side and travel with half the speed of ball 1 as in the figure?



Example 9.4:
The Executive Stress Reliever

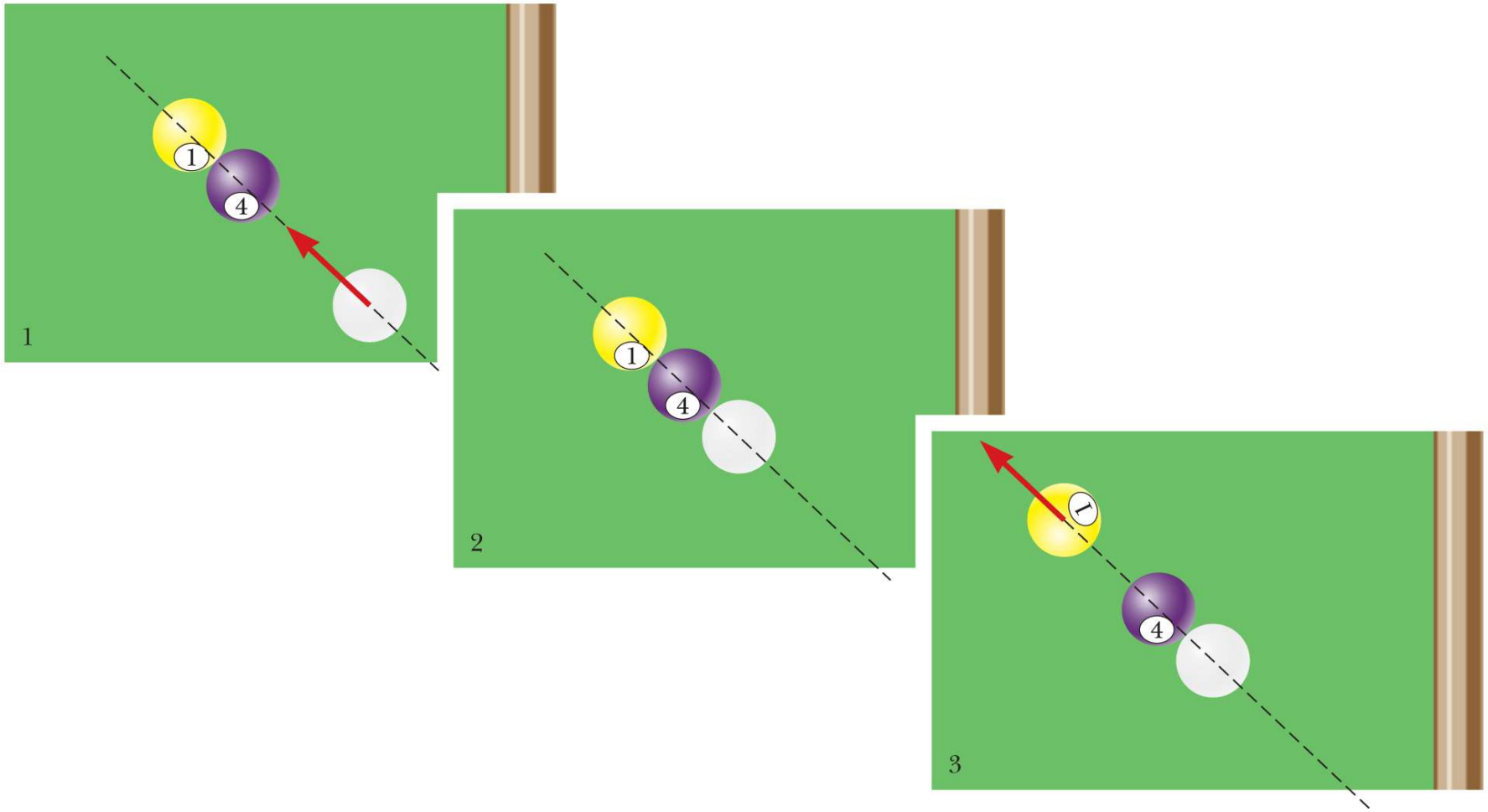
$$m\left(\frac{v}{2}\right)+m\left(\frac{v}{2}\right)=mv$$



$$K_i=\frac{1}{2}mv^2$$

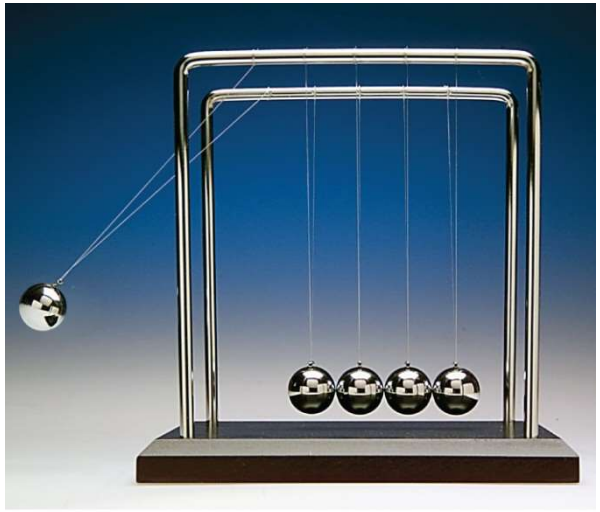
$$K_f=\frac{1}{2}m\left(\frac{v}{2}\right)^2+\frac{1}{2}m\left(\frac{v}{2}\right)^2=\frac{1}{4}mv^2$$

Example 9.4:
The Executive Stress Reliever



Example 9.4:
The Executive Stress Reliever

Consider what would happen if balls 4 and 5 are glued together. Now what happens when ball 1 is pulled out and released?



a

Example 9.4:
The Executive Stress Reliever

$$p_i = p_f$$

$$mv_{1i} = mv_{1f} + 2mv_{4,5}$$

$$K_i = K_f$$

$$\frac{1}{2}mv_{1i}^{\,2} = \frac{1}{2}mv_{1f}^{\,2} + \frac{1}{2}(2m)v_{4,5}^{\,2}$$

$$v_{4,5} = \frac{2}{3}v_{1i} \qquad v_{1f} = -\frac{1}{3}v_{1i}$$

Example 9.5:
Carry Collision Insurance!

An 1 800-kg car stopped at a traffic light is struck from the rear by a 900-kg car. The two cars become entangled, moving along the same path as that of the originally moving car. If the smaller car were moving at 20.0 m/s before the collision, what is the velocity of the entangled cars after the collision?

Example 9.5:
Carry Collision Insurance!

Inelastic

$$\Delta\vec{\mathbf{p}} = 0 \Rightarrow p_i = p_f \Rightarrow m_1 v_1 = \left(m_1 + m_2\right) v_f$$

20 m/s

$$v_f = \frac{m_1 v_1}{m_1 + m_2} = \frac{\left(900\ \text{kg}\right)\left(20.0\ \text{m/s}\right)}{900\ \text{kg} + 1800\ \text{kg}} = \boxed{6.67\ \text{m/s}}$$

Example 9.5:
Carry Collision Insurance!

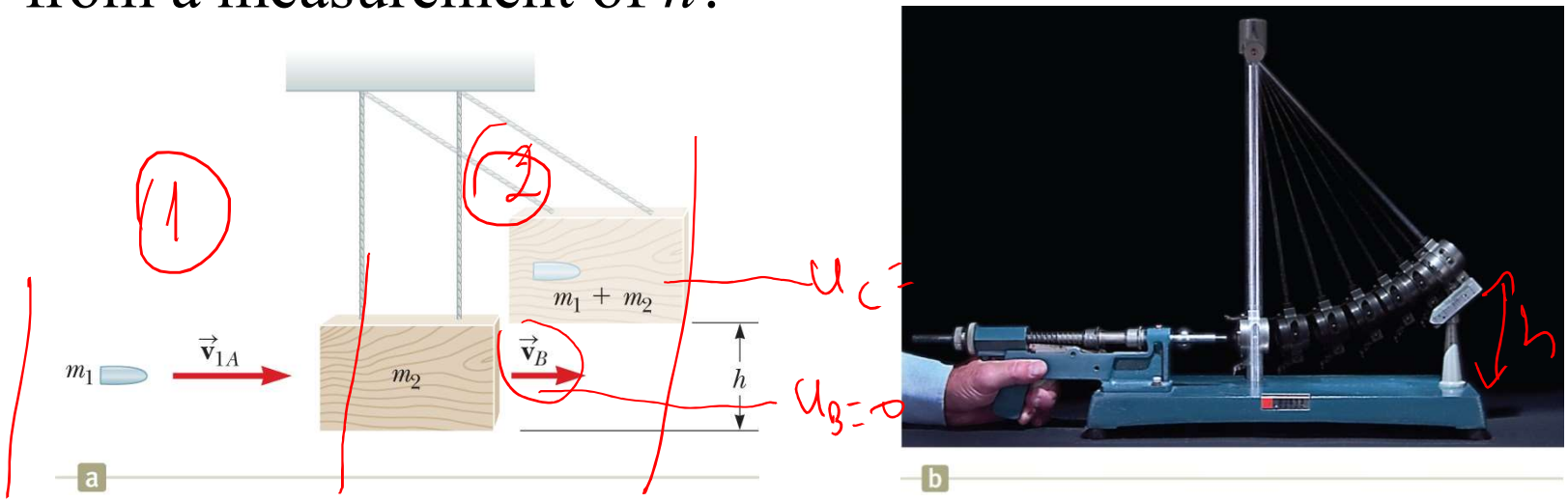
Suppose we exchange the masses of the cars. What if a stationary 900-kg car is struck by a moving 1 800-kg car? Is the final speed the same as before?

Final speed is higher.

$$v_f = \frac{m_1 v_1}{m_1 + m_2} = \frac{(1800\ \text{kg})(20.0\ \text{m/s})}{1800\ \text{kg} + 900\ \text{kg}} = \boxed{13.3\ \text{m/s}}$$

Example 9.6:
The Ballistic Pendulum

The ballistic pendulum an apparatus used to measure the speed of a fast-moving projectile such as a bullet. A projectile of mass m_1 is fired into a large block of wood of mass m_2 suspended from some light wires. The projectile embeds in the block, and the entire system swings through a height h. How can we determine the speed of the projectile from a measurement of h?



Example 9.6:
The Ballistic Pendulum

$$\vec{\mathbf{v}}_B = \frac{m_1\vec{\mathbf{v}}_{1A} + m_2\vec{\mathbf{v}}_{2A}}{m_1 + m_2} \Rightarrow v_B = \frac{m_1 v_{1A}}{m_1 + m_2}$$

$$K_B = \frac{1}{2}\left(m_1 + m_2\right)v_B^2 \Rightarrow K_B = \frac{m_1^{\,2} v_{1A}^{\,2}}{2\left(m_1 + m_2\right)}$$

$$U_B = 0 \text{ and } U_C = \left(m_1 + m_2\right)gh$$

$$\Delta K + \Delta U = 0 \rightarrow \left(K_C - K_B\right) + \left(U_C - U_B\right) = 0$$

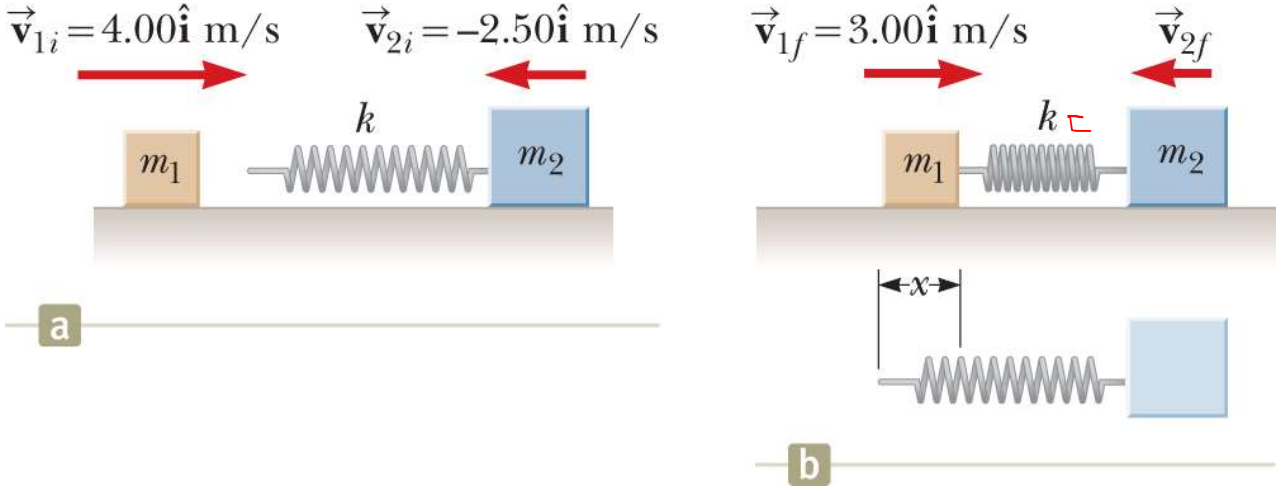
$$\left[0 - \frac{m_1^{\,2} v_{1A}^{\,2}}{2\left(m_1 + m_2\right)}\right] + \left[\left(m_1 + m_2\right)gh - 0\right] = 0$$

$$v_{1A} = \left(\frac{m_1 + m_2}{m_1}\right)\sqrt{2gh}$$

Example 9.7:
A Two-Body Collision with a Spring

A block of mass m_1 = 1.60 kg initially moving to the right with a speed of 4.00 m/s on a frictionless, horizontal track collides with a light spring attached to a second block of mass m_2 = 2.10 kg initially moving to the left with a speed of 2.50 m/s as shown in the figure on the left. The spring constant is 600 N/m.

(**A**) Find the velocities of the two blocks when they are again moving separately after the collision.



Example 9.7:
A Two-Body Collision with a Spring

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (1)$$

since collision elastic

$$v_{1i} - v_{21} = -\left(v_{1f} - v_{2f}\right) \quad (2)$$

$$m_1 v_{1i} - m_1 v_{2i} = -m_1 v_{1f} + m_1 v_{2f} \quad (3)$$

(1) + (3)

$$2m_1 v_{1i} + \left(m_2 - m_1\right) v_{2i} = \left(m_1 + m_2\right) v_{2f}$$

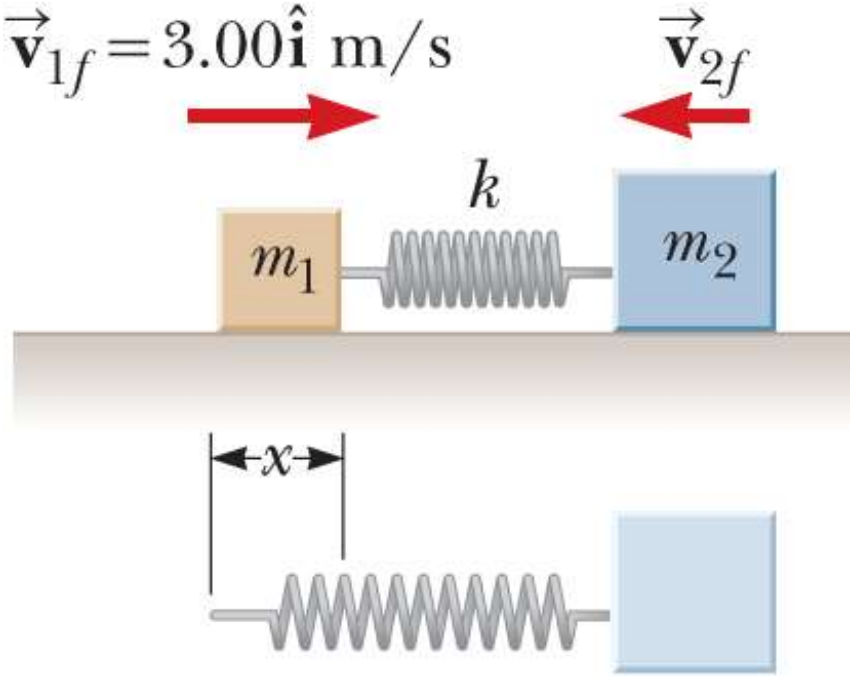
$$v_{2f} = \frac{2m_1 v_{1i} + \left(m_2 - m_1\right) v_{2i}}{m_1 + m_2}$$

$$v_{2f} = \frac{2(1.60\ \text{kg})(4.00\ \text{m/s}) + (2.10\ \text{kg} - 1.60\ \text{kg})(-2.50\ \text{m/s})}{1.60\ \text{kg} + 2.10\ \text{kg}} = 3.12\ \text{m/s}$$

$$v_{1f} = v_{2f} - v_{1i} + v_{2i} = 3.12\ \text{m/s} - 4.00\ \text{m/s} + (-2.50\ \text{m/s}) = \boxed{-3.38\ \text{m/s}}$$

Example 9.7:
A Two-Body Collision with a Spring

(**B**) Determine the velocity of block 2 during the collision, at the instant block 1 is moving to the right with a velocity of +3.00 m/s as in the figure.



Example 9.7:
A Two-Body Collision with a Spring

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

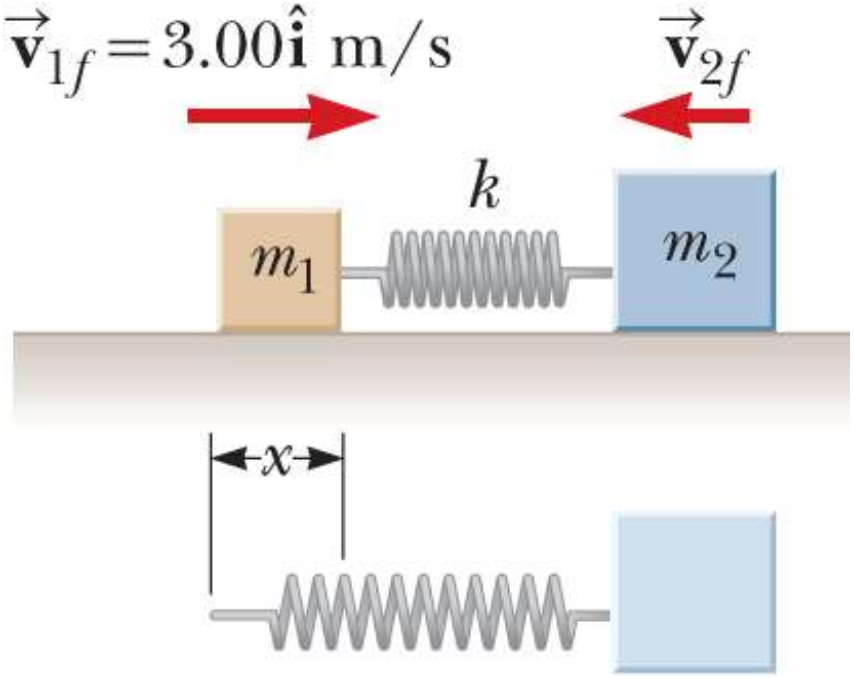
$$v_{2f} = \frac{m_1 v_{1i} + m_2 v_{2i} - m_1 v_{1f}}{m_2}$$

$$v_{2f} = \frac{(1.60\ \text{kg})(4.00\ \text{m/s}) + (2.10\ \text{kg})(-2.50\ \text{m/s}) - (1.60\ \text{kg})(3.00\ \text{m/s})}{2.10\ \text{kg}}$$

$$= \boxed{-1.74\ \text{m/s}}$$

Example 9.7:
A Two-Body Collision with a Spring

(**C**) Determine the distance the spring is compressed at that instant.



Example 9.7:
A Two-Body Collision with a Spring

$$\Delta K + \Delta U = 0$$

$$\left[\left(\frac{1}{2}m_1 v_{1f}^{\,2} + \frac{1}{2}m_2 v_{2f}^{\,2}\right) - \left(\frac{1}{2}m_1 v_{1i}^{\,2} + \frac{1}{2}m_2 v_{2i}^{\,2}\right)\right] + \left(\frac{1}{2}kx^2 - 0\right) = 0$$

$$x^2 = \frac{1}{k}\left[m_1\left(v_{1i}^{\,2} - v_{1f}^{\,2}\right) + m_2\left(v_{2i}^{\,2} - v_{2f}^{\,2}\right)\right]$$

$$x^2 = \left(\frac{1}{600\ \text{N/m}}\right)$$

$$\times\left\{(1.60\ \text{kg})\left[(4.00\ \text{m/s})^2 - (3.00\ \text{m/s})^2\right] + (2.10\ \text{kg})\left[(2.50\ \text{m/s})^2 - (1.74\ \text{m/s})^2\right]\right\}$$

$$\Rightarrow x = \boxed{0.173\ \text{m}}$$

Collisions in Two Dimensions

$$m_1 v_{1x} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

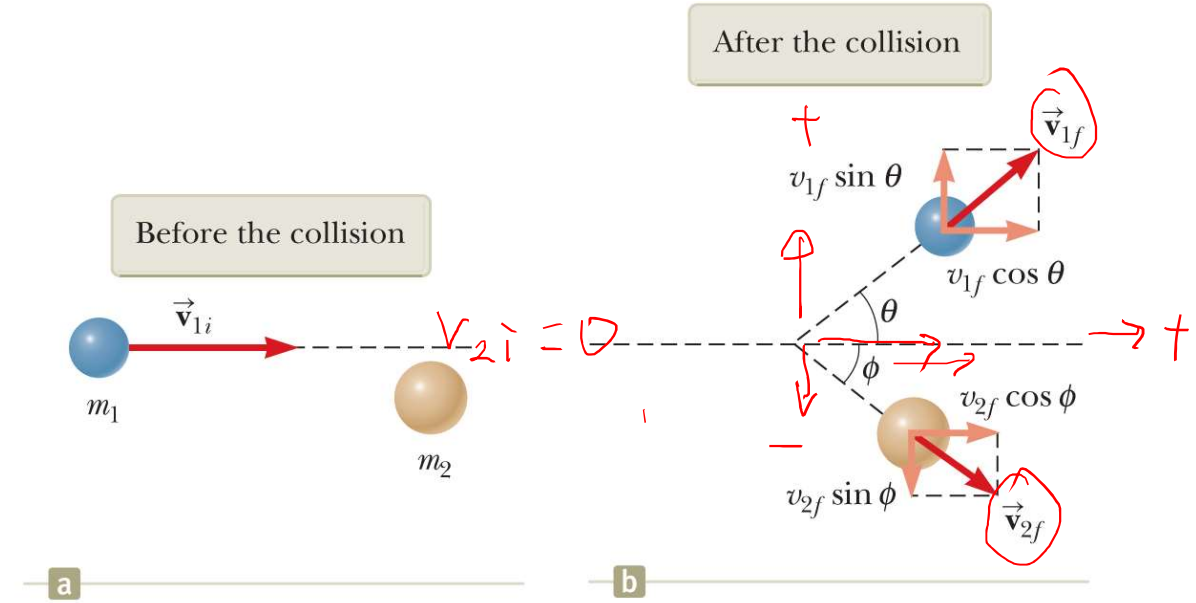
$$m_1 v_{1y} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

Collisions in Two Dimensions

$$\Delta p_x = 0 \Rightarrow p_{xi} = p_{xf} \Rightarrow m_1 v_{1i} = m_1 v_{1f} \cos\theta + m_2 v_{2f} \cos\phi$$

$$\Delta p_y = 0 \Rightarrow p_{yi} = p_{yf} \Rightarrow 0 = m_1 v_{1f} \sin\theta - m_2 v_{2f} \sin\phi$$

$$K_i = K_f \Rightarrow \frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$



Problem-Solving Strategy: Two-Dimensional Collisions

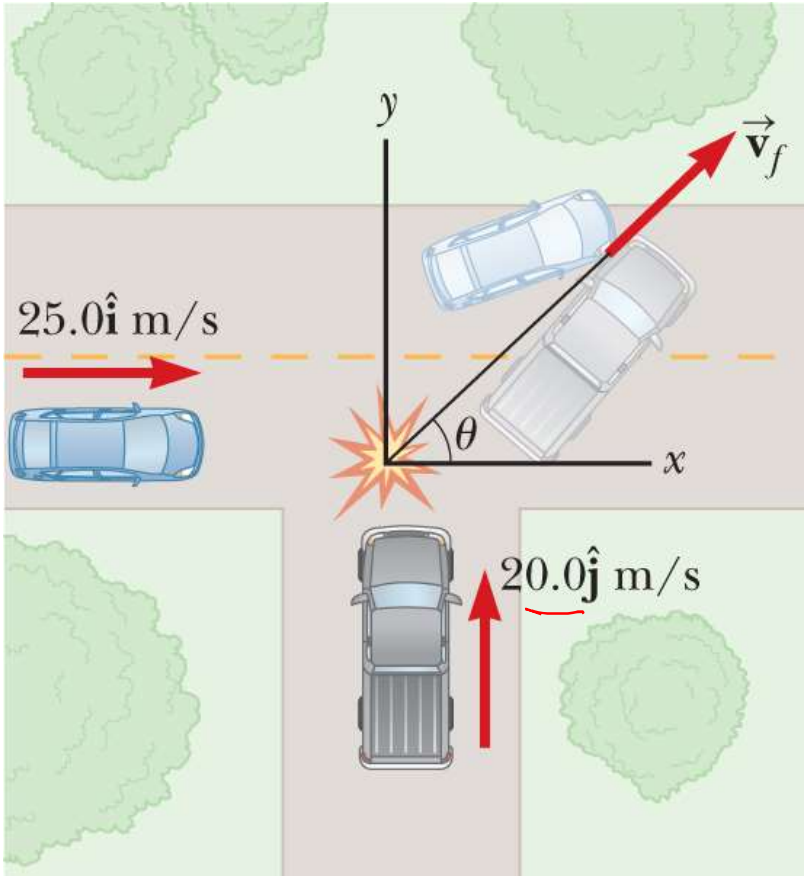
1. Conceptualize
2. Categorize: elastic? perfectly inelastic? inelastic?
3. Analyze

$$\Delta\vec{\mathbf{p}} = 0 \Rightarrow p_{ix} = p_{fx} \text{ and } p_{iy} = p_{fy}$$

4. Finalize

Example 9.8:
Collisions at an Intersection

A 1 500-kg car traveling east with a speed of 25.0 m/s collides at an intersection with a 2 500-kg truck traveling north at a speed of 20.0 m/s as shown in the figure. Find the direction and magnitude of the velocity of the wreckage after the collision, assuming the vehicles stick together after the collision.



Example 9.8: Collisions at an Intersection

$$\Delta p_x = 0 \Rightarrow \sum p_{xi} = \sum p_{xf} \Rightarrow m_1 v_{1i} = (m_1 + m_2) v_f \cos\theta \quad (1)$$

$$\Delta p_y = 0 \Rightarrow \sum p_{yi} = \sum p_{yf} \Rightarrow m_2 v_{2i} = (m_1 + m_2) v_f \sin\theta \quad (2)$$

$$\frac{(2)}{(1)} \qquad \frac{m_2 v_{2i}}{m_1 v_{1i}} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

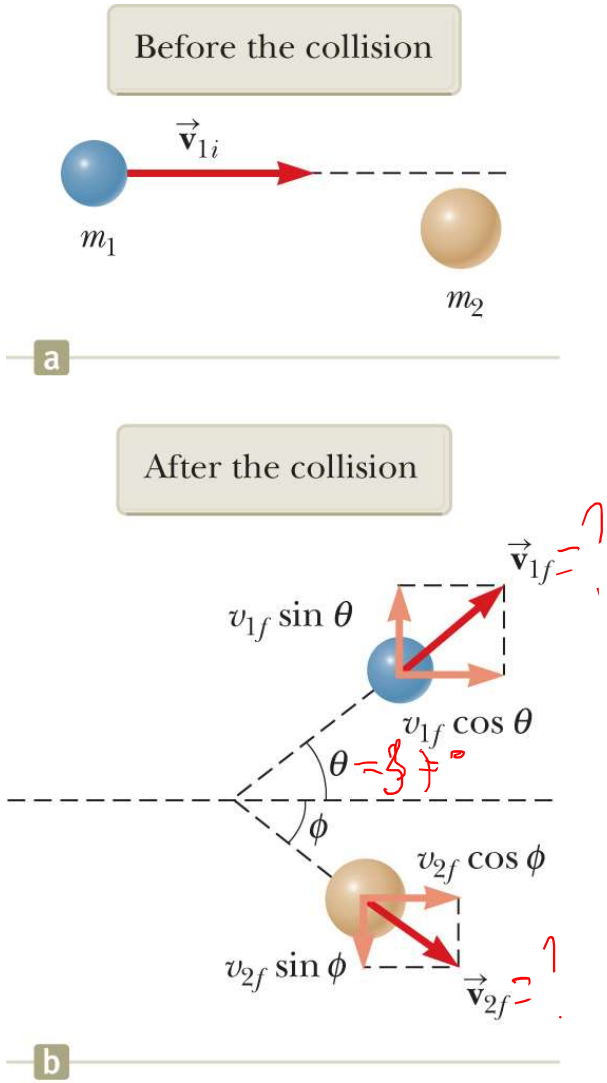
$$\theta = \tan^{-1}\left(\frac{m_2 v_{2i}}{m_1 v_{1i}}\right) = \tan^{-1}\left[\frac{(2500\ \text{kg})(20.0\ \text{m/s})}{(1500\ \text{kg})(25.0\ \text{m/s})}\right] = \boxed{53.1^\circ}$$

direction

$$v_f = \frac{m_2 v_{2i}}{(m_1 + m_2)\sin\theta} = \frac{(2500\ \text{kg})(20.0\ \text{m/s})}{(1500\ \text{kg} + 2500\ \text{kg})\sin 53.1^\circ} = \boxed{15.6\ \text{m/s}}$$

Example 9.9:
Proton-Proton Collision

A proton collides elastically with another proton that is initially at rest. The incoming proton has an initial speed of 3.50×10^5 m/s and makes a glancing collision with the second proton as in shown in the figure (At close separations, the protons exert a repulsive electrostatic force on each other.) After the collision, one proton moves off at an angle of 37.0° to the original direction of motion and the second deflects at an angle of ϕ to the same axis. Find the final speeds of the two protons and the angle ϕ.



Example 9.9: Proton-Proton Collision

$$(1)\quad v_{1i} = v_{1f}\cos\theta + v_{2f}\cos\phi$$

$$(2)\quad 0 = v_{1f}\sin\theta - v_{2f}\sin\phi$$

$$(3)\quad v_{1i}^{\,2} = v_{1f}^{\,2} + v_{2f}^{\,2}$$

$$v_{2f}\cos\phi = v_{1i} - v_{1f}\cos\theta$$

$$v_{2f}\sin\phi = v_{1f}\sin\theta$$

$$v_{2f}^{\,2}\cos^2\phi + v_{2f}^{\,2}\sin^2\phi = v_{1i}^{\,2} - 2v_{1i}v_{1f}\cos\theta + v_{1f}^{\,2}\cos^2\theta + v_{1f}^{\,2}\sin^2\theta$$

$$(4)\quad v_{2f}^{\,2} = v_{1i}^{\,2} - 2v_{1i}v_{1f}\cos\theta + v_{1f}^{\,2}$$

Example 9.9: Proton-Proton Collision

substitute equation (4) into equation (3)

$$v_{1f}^{\,2} + \left(v_{1i}^{\,2} - 2v_{1i}v_{1f}\cos\theta + v_{1f}^{\,2}\right) = v_{1i}^{\,2}$$

$$(5)\quad v_{1f}^{\,2} - v_{1i}v_{1f}\cos\theta = 0 \Rightarrow v_{1f}\left(v_{1f} - v_{1i}\cos\theta\right) = 0$$

$v_{1f} = 0$

$$v_{1f} = v_{1i}\cos\theta = \left(3.50\times10^{5}\ \text{m/s}\right)\cos 37.0^\circ$$

$$v_{1f} = \boxed{2.80\times10^{5}\ \text{m/s}}$$

(3)

$$v_{2f} = \sqrt{v_{1i}^{\,2} - v_{1f}^{\,2}} = \sqrt{\left(3.50\times10^{5}\ \text{m/s}\right)^2 - \left(2.80\times10^{5}\ \text{m/s}\right)^2}$$

$$= \boxed{2.11\times10^{5}\ \text{m/s}}$$

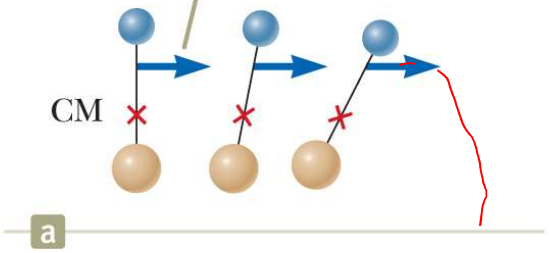
(4)

$$\phi = \sin^{-1}\left(\frac{v_{1f}\sin\theta}{v_{2f}}\right) = \sin^{-1}\left[\frac{\left(2.80\times10^{5}\ \text{m/s}\right)\sin 37.0^\circ}{\left(2.11\times10^{5}\ \text{m/s}\right)}\right] = \boxed{53.0^\circ}$$

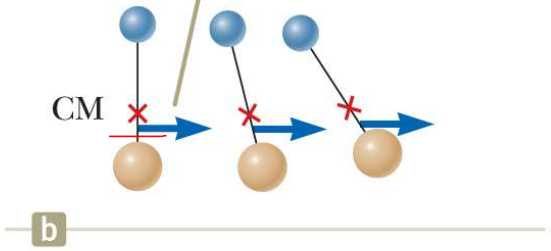
The Center of Mass

translational motion of CM

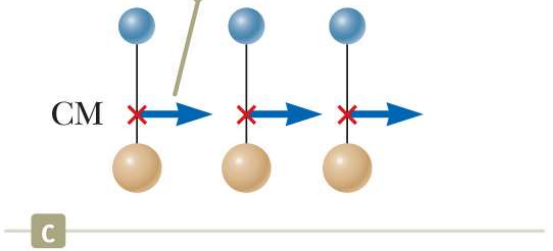
The system rotates clockwise when a force is applied above the center of mass.



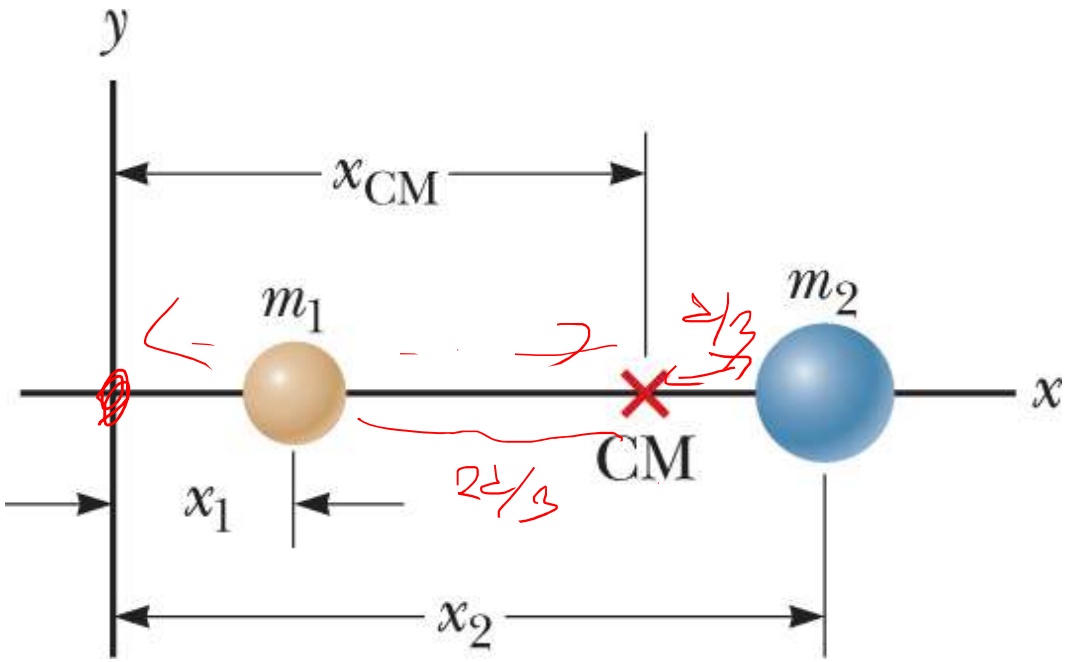
The system rotates counter-clockwise when a force is applied below the center of mass.



The system moves in the direction of the force without rotating when a force is applied at the center of mass.



The Center of Mass



$$x_{CM} \equiv \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

mass*distance to reference point / total mass

$$\text{Let } x_1 = 0,\ x_2 = d,\ m_2 = 2m_1$$

$$x_{CM} = \frac{m_1(0) + (2m_1)d}{m_1 + 2m_1} = \frac{2d}{3}$$

The Center of Mass

many particles with masses m_i

$$x_{\mathrm{CM}} \equiv \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \ldots + m_n x_n}{m_1 + m_2 + m_3 + \ldots + m_n}$$

$$X_{CM} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{\sum_i m_i x_i}{M} = \frac{1}{M}\sum_i m_i x_i$$

$$y_{\mathrm{CM}} \equiv \frac{1}{M}\sum_i m_i y_i \quad \text{and} \quad z_{\mathrm{CM}} \equiv \frac{1}{M}\sum_i m_i z_i$$

The Center of Mass

Center of mass in 3D.

$$\vec{\mathbf{r}}_{\mathrm{CM}} = x_{\mathrm{CM}}\hat{\mathbf{i}} + y_{\mathrm{CM}}\hat{\mathbf{j}} + z_{\mathrm{CM}}\hat{\mathbf{k}}$$ → position vector

$$= \frac{1}{M}\sum_i m_i x_i \hat{\mathbf{i}} + \frac{1}{M}\sum_i m_i y_i \hat{\mathbf{j}} + \frac{1}{M}\sum_i m_i z_i \hat{\mathbf{k}}$$

$$\vec{\mathbf{r}}_{\mathrm{CM}} \equiv \frac{1}{M}\sum_i m_i \vec{\mathbf{r}}_i \qquad \rightarrow \vec{\mathbf{r}}_i \equiv x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}} + z_i\hat{\mathbf{k}}$$

The Center of Mass

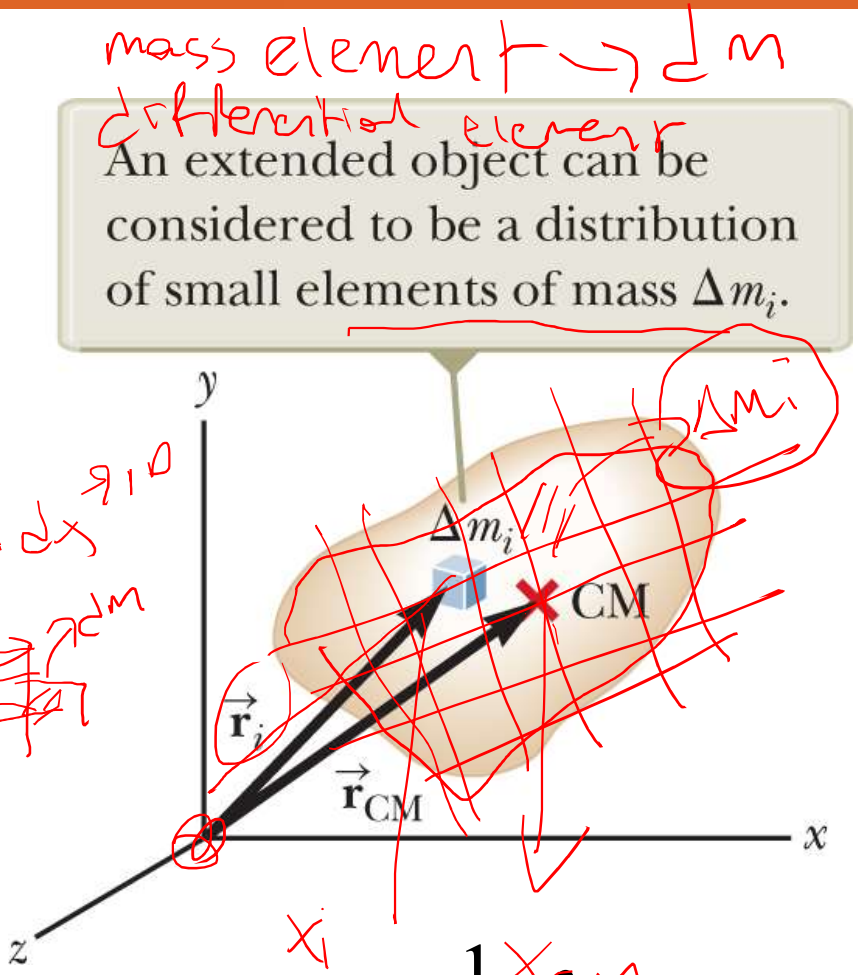
$$x_{\text{CM}} \approx \frac{1}{M}\sum_i x_i \Delta m_i$$

$$x_{\text{CM}} = \lim_{\Delta m_i \to 0} \frac{1}{M}\sum_i x_i \Delta m_i$$

$$x_{\text{CM}} = \frac{1}{M}\int x\,dm$$

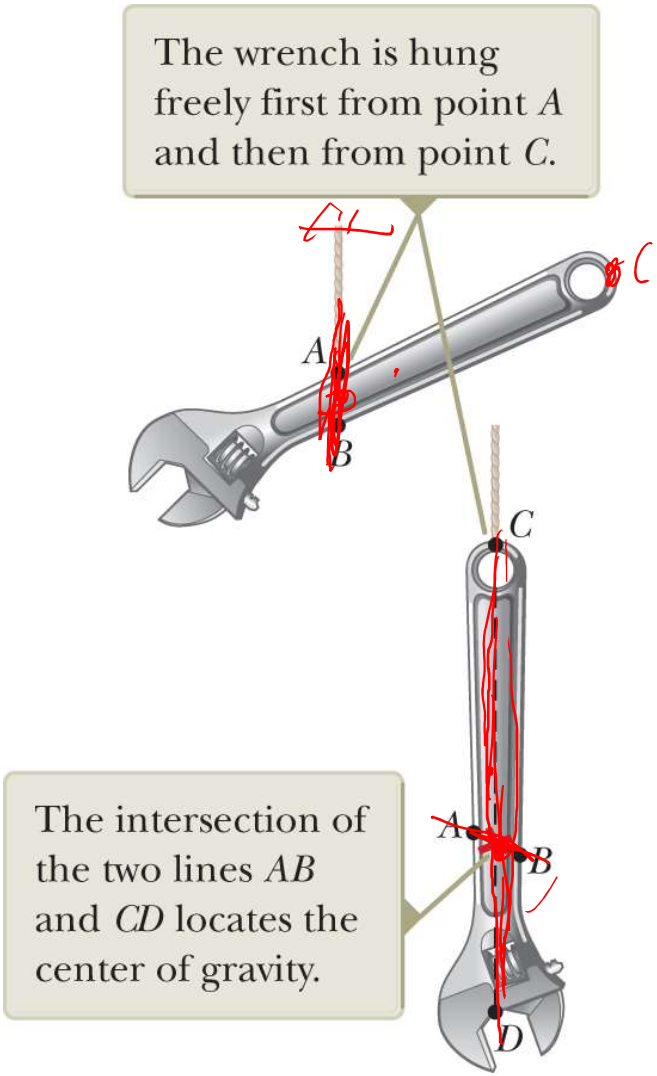
$$y_{\text{CM}} = \frac{1}{M}\int y\,dm$$

and $$z_{\text{CM}} = \frac{1}{M}\int z\,dm$$



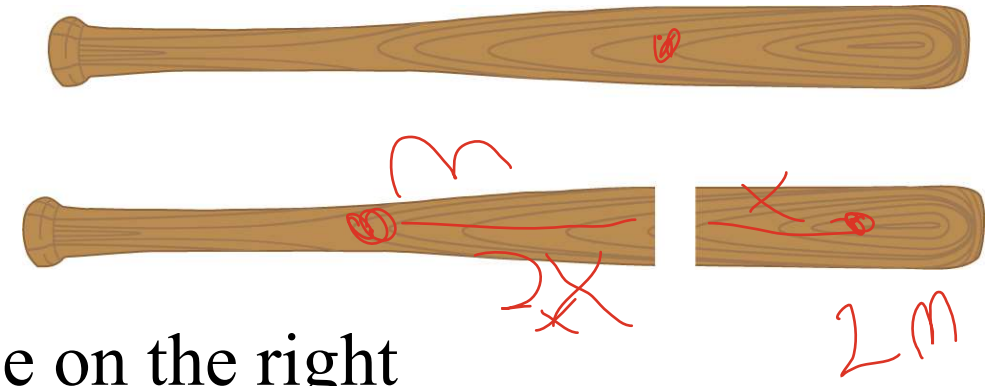
$$\vec{\mathbf{r}}_{\text{CM}} = \frac{1}{M}\int \vec{\mathbf{r}}\,dm$$

The Center of Gravity



Quick Quiz 9.7

A baseball bat of uniform density is cut at the location of its center of mass as shown in the figure. Which piece has the smaller mass?



(a) the piece on the right

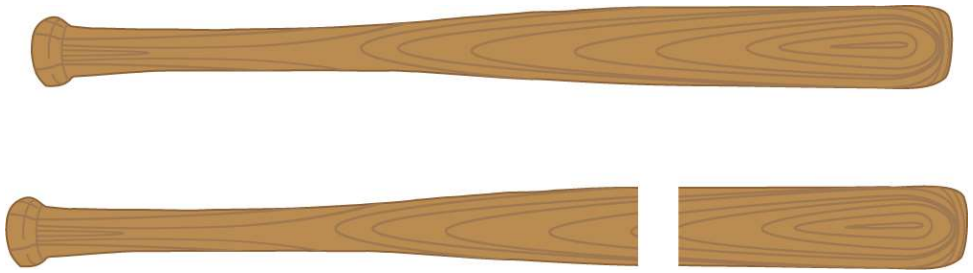
(b) the piece on the left

(c) both pieces have the same mass

(d) impossible to determine

Quick Quiz 9.7

A baseball bat of uniform density is cut at the location of its center of mass as shown in the figure. Which piece has the smaller mass?



(a) the piece on the right

(b) the piece on the left

(c) both pieces have the same mass

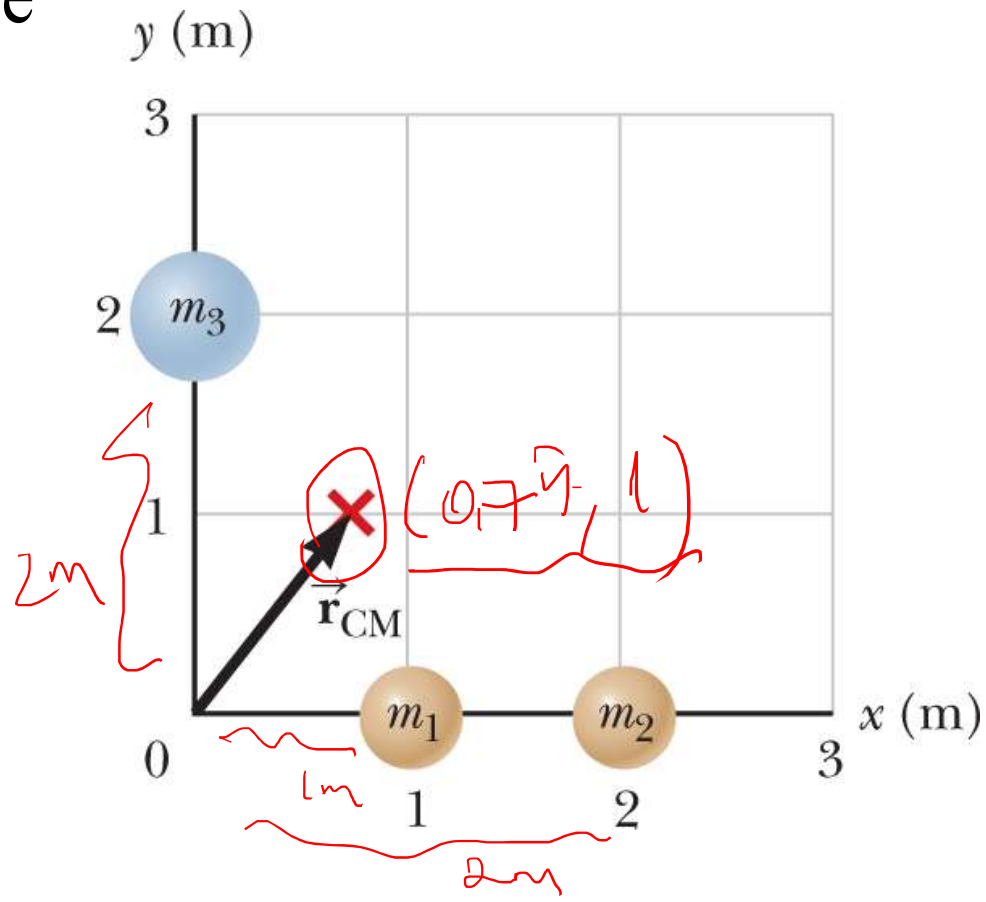
(d) impossible to determine

Example 9.10:
The Center of Mass of Three Particles

A system consists of three particles located as shown in the figure. Find the center of mass of the system. The masses of the particles are

$m_1 = m_2 = 1.0$ kg and

$m_3 = 2.0$ kg.



Example 9.10:
The Center of Mass of Three Particles

$$x_{\text{CM}} = \frac{1}{M}\sum_i m_i x_i = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$= \frac{(1.0\ \text{kg})(1.0\ \text{m}) + (1.0\ \text{kg})(2.0\ \text{m}) + (2.0\ \text{kg})(0)}{1.0\ \text{kg} + 1.0\ \text{kg} + 2.0\ \text{kg}}$$

$$= \frac{3.0\ \text{kg}\cdot\text{m}}{4.0\ \text{kg}} = 0.75\ \text{m}$$

$$y_{\text{CM}} = \frac{1}{M}\sum_i m_i y_i = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3}$$

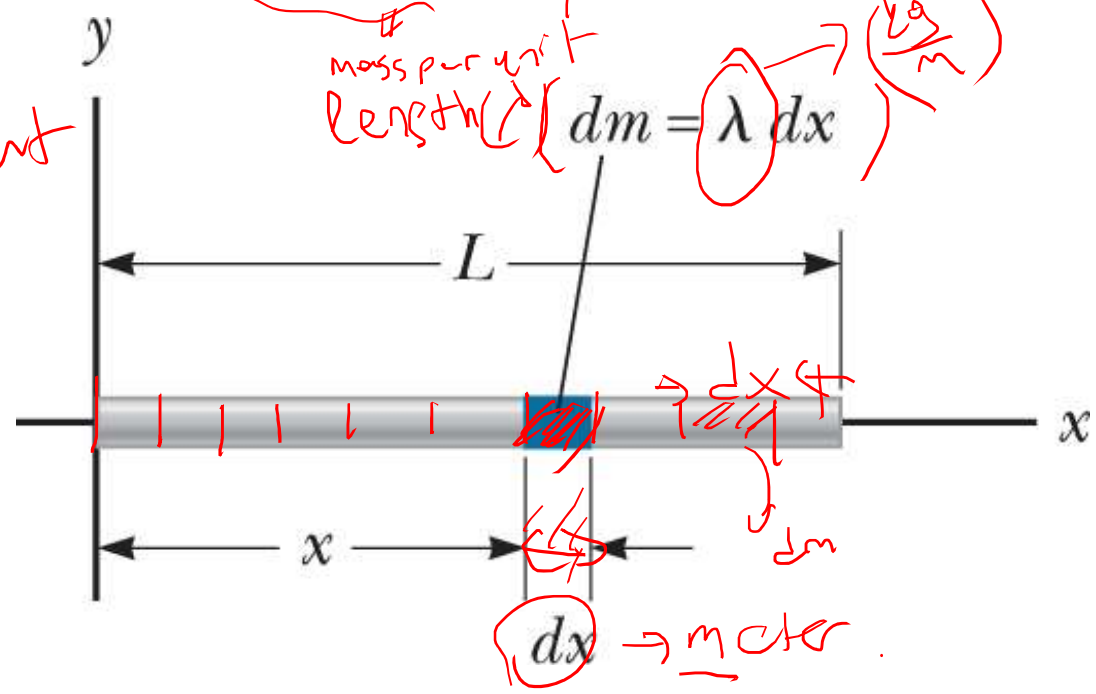
$$= \frac{(1.0\ \text{kg})(0) + (1.0\ \text{kg})(0) + (2.0\ \text{kg})(2.0\ \text{m})}{1.0\ \text{kg} + 1.0\ \text{kg} + 2.0\ \text{kg}}$$

$$= \frac{4.0\ \text{kg}\cdot\text{m}}{4.0\ \text{kg}} = 1.0\ \text{m}$$

$$r_{\text{CM}} \equiv x_{\text{CM}}\hat{\mathbf{i}} + y_{\text{CM}}\hat{\mathbf{j}} = \boxed{\left(0.75\hat{\mathbf{i}} + 1.0\hat{\mathbf{j}}\right)\ \text{m}}$$

Example 9.11: The Center of Mass of a Rod

(**A**) Show that the center of mass of a rod of mass M and length L lies midway between its ends, assuming the rod has a uniform mass per unit length.

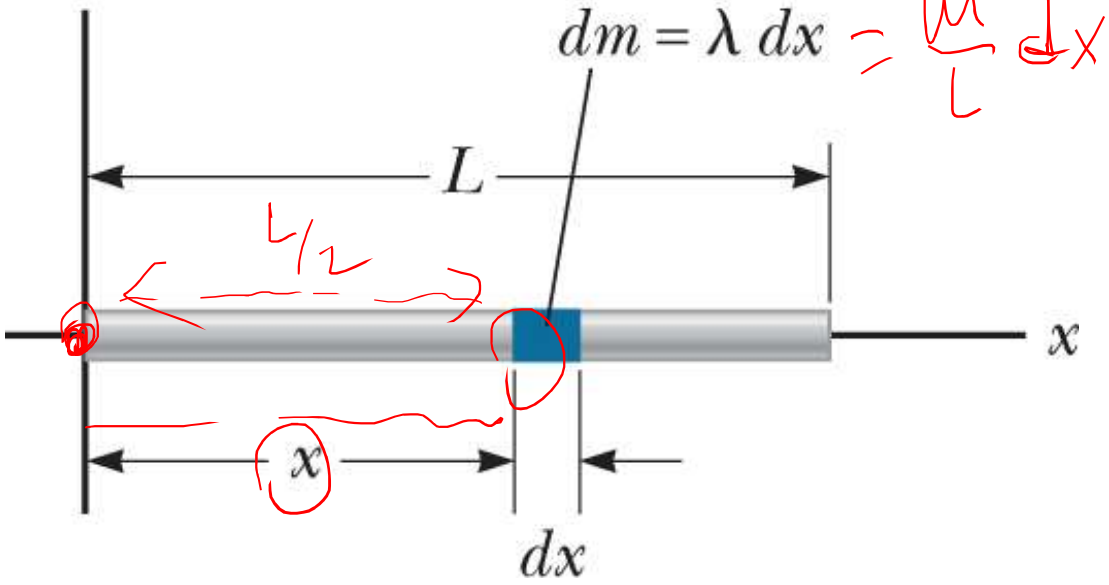


Example 9.11: The Center of Mass of a Rod

$$x_{\text{CM}} = \frac{1}{M}\int x\,dm = \int_0^L x\lambda\,dx = \frac{\lambda}{M}\frac{x^2}{2}\Big|_0^L = \frac{\lambda L^2}{2M}$$

$$x_{\text{CM}} = \frac{L^2}{2M}\left(\frac{M}{L}\right)$$

$$x_{\text{CM}} = \frac{1}{2}L$$



Example 9.11:
The Center of Mass of a Rod

(**B**) Suppose a rod is *nonuniform* such that its mass per unit length varies linearly with x according to the expression $\lambda = \alpha x$, where α is a constant. Find the x coordinate of the center of mass as a fraction of L.

Example 9.11: The Center of Mass of a Rod

$$dm = \lambda\, dx, \quad \lambda = \alpha x$$

$$x_{\text{CM}} = \frac{1}{M}\int x\, dm = \frac{1}{M}\int_0^L x\lambda\, dx$$

$$= \frac{1}{M}\int_0^L x\alpha x\, dx = \frac{\alpha}{M}\int_0^L x^2\, dx = \frac{\alpha L^3}{3M}$$

$$\rightarrow M = \int dm = \int_0^L \lambda\, dx = \int_0^L \alpha x\, dx = \frac{\alpha L^2}{2}$$

$$x_{\text{CM}} = \frac{\alpha L^3}{3\alpha L^2/2} = \frac{2}{3}L$$

2D object

$$x_{cm} = \frac{\int x\,dm}{M}$$

$dm = \left(\frac{M}{A}\right) dA$ → mass of an area element = dA

$\frac{M}{A}$ = mass per unit area

$dm = \frac{M}{ab} \cdot y \cdot dx$ ← dm

Area of triangle $= \frac{ab}{2}$

$= \frac{2M}{ab} \cdot \frac{b}{a} x\,dx$

y

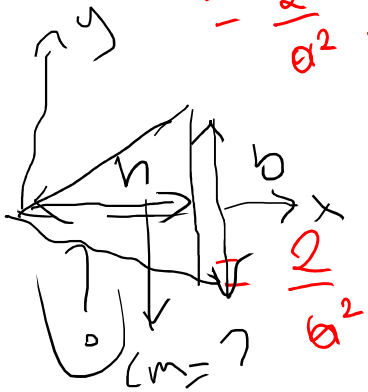
$$dm = \frac{2M}{a^2} x\,dx$$

$$x_{cm} = \frac{\int x \cdot \frac{2M}{a^2} x\,dx}{M}$$

$$= \frac{\frac{2M}{a^2}}{M} \int_0^a x^2\,dx$$

$$= \frac{2}{a^2} \int_0^a x^2\,dx$$

$$\frac{x^3}{3}\Big|_0^a$$

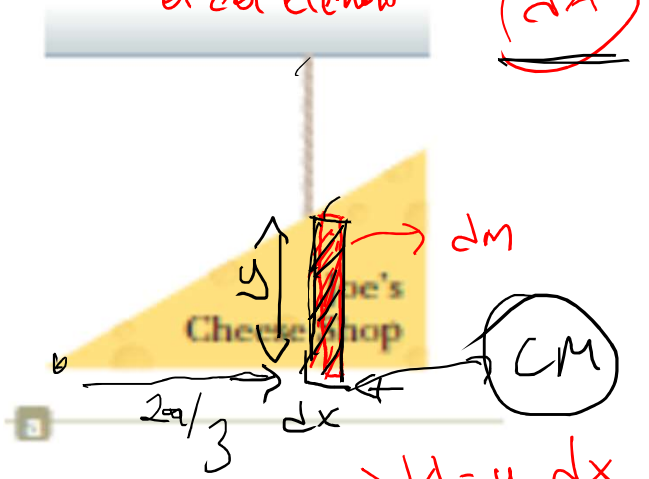


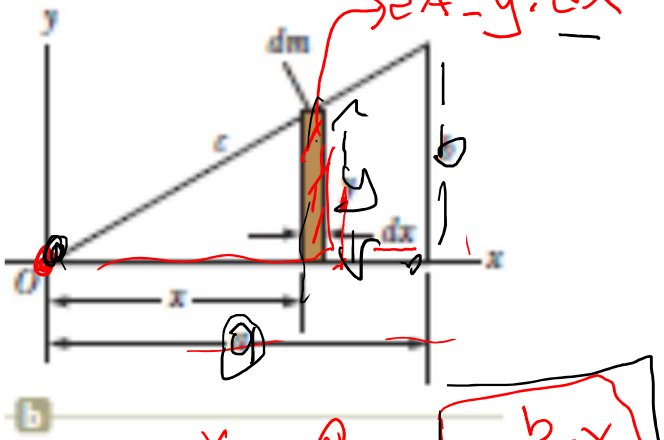
$$\frac{2}{a^2} \cdot \frac{a^3}{3} = \frac{2a}{3} = x_{cm}$$



$2\pi r$ → $2\pi r \cdot dr$

$dA = 2\pi r\,dr$





$$\frac{x}{y} = \frac{a}{b} \Rightarrow y = \frac{b}{a} x$$

Systems of Many Particles

velocity of CM.

$$\vec{\mathbf{v}}_{\mathrm{CM}} = \frac{d\vec{\mathbf{r}}_{\mathrm{CM}}}{dt} = \frac{1}{M}\sum_i m_i \frac{d\vec{\mathbf{r}}_i}{dt} = \frac{1}{M}\sum_i m_i \vec{\mathbf{v}}_i$$

total momentum of particles.

$$M\vec{\mathbf{v}}_{\mathrm{CM}} = \sum_i m_i \vec{\mathbf{v}}_i = \sum_i \vec{\mathbf{p}}_i = \vec{\mathbf{p}}_{\mathrm{tot}}$$

$$r_{CM} = \frac{\sum m_i r_i}{\sum m_i} = \frac{\sum m_i r_i}{M}$$

acceleration of CM.

$$\vec{\mathbf{a}}_{\mathrm{CM}} = \frac{d\vec{\mathbf{v}}_{\mathrm{CM}}}{dt} = \frac{1}{M}\sum_i m_i \frac{d\vec{\mathbf{v}}}{dt} = \frac{1}{M}\sum_i m_i \vec{\mathbf{a}}$$

$$\frac{d}{dt}\left(\frac{\sum m_i r_i}{M}\right)$$

$F_i = F_{ext} + F_{int}$

$\sum F_{int} = 0$

$$M\vec{\mathbf{a}}_{\mathrm{CM}} = \sum_i m_i \vec{\mathbf{a}} = \sum_i \vec{\mathbf{F}}_i$$

$$\boxed{\sum_i \vec{\mathbf{F}}_{\mathrm{ext}} = M\vec{\mathbf{a}}_{\mathrm{CM}}}$$

Net force on each particle

$\frac{\sum m_i}{M}$ $\frac{d\vec{r}_i}{dt}$ v_i

Systems of Many Particles

The center of mass of a system of particles having combined mass M moves like an equivalent single particle of mass M would move under the influence of the net external force on the system.

$$\int \sum \vec{\mathbf{F}}_{\text{ext}}\, dt = \int M\vec{\mathbf{a}}_{\text{CM}}\, dt = \int M \frac{d\,\vec{\mathbf{v}}_{\text{CM}}}{dt} dt = M \int \frac{d\,\vec{\mathbf{v}}_{\text{CM}}}{dt} dt = M\,\Delta\vec{\mathbf{v}}_{\text{CM}}$$

$$\Delta\vec{\mathbf{p}}_{\text{tot}} = \vec{\mathbf{I}}$$ → impulse-momentum theorem

If $\vec{F}_{ext} = 0$ $\quad M\vec{\mathbf{a}}_{\text{CM}} = M \frac{d\vec{\mathbf{v}}_{\text{CM}}}{dt} = 0$ $\quad \frac{d}{dt}(MV_{cm}) = 0$

$$\Delta\vec{\mathbf{p}}_{\text{tot}} = 0 \qquad M\vec{\mathbf{v}}_{\text{CM}} = \vec{\mathbf{p}}_{\text{tot}} = \text{constant} \left(\text{when } \sum \vec{\mathbf{F}}_{\text{ext}} = 0\right)$$

Quick Quiz 9.8 Part I

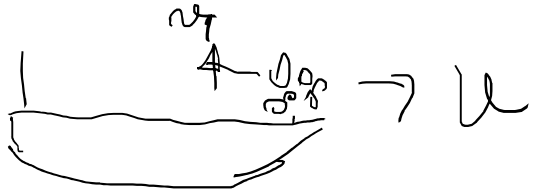
A cruise ship is moving at constant speed through the water. The vacationers on the ship are eager to arrive at their next destination. They decide to try to speed up the cruise ship by gathering at the bow (the front) and running together toward the stern (the back) of the ship. While they are running toward the stern, the speed of the ship is

(a) higher than it was before.

(b) unchanged.

(c) lower than it was before.

(d) impossible to determine.



$$\sum F_{ext} = 0$$

$$P_{tot} = \text{constant}$$

Quick Quiz 9.8 Part I

A cruise ship is moving at constant speed through the water. The vacationers on the ship are eager to arrive at their next destination. They decide to try to speed up the cruise ship by gathering at the bow (the front) and running together toward the stern (the back) of the ship. While they are running toward the stern, the speed of the ship is

(a) higher than it was before.

(b) unchanged.

(c) lower than it was before.

(d) impossible to determine.



Quick Quiz 9.8 Part II

A cruise ship is moving at constant speed through the water. The vacationers on the ship are eager to arrive at their next destination. They decide to try to speed up the cruise ship by gathering at the bow (the front) and running together toward the stern (the back) of the ship. The vacationers stop running when they reach the stern of the ship. After they have all stopped running, the speed of the ship is

(a) higher than it was before they started running.

(b) unchanged from what it was before they started running.

(c) lower than it was before they started running.

(d) impossible to determine.

Quick Quiz 9.8 Part II

A cruise ship is moving at constant speed through the water. The vacationers on the ship are eager to arrive at their next destination. They decide to try to speed up the cruise ship by gathering at the bow (the front) and running together toward the stern (the back) of the ship. The vacationers stop running when they reach the stern of the ship. After they have all stopped running, the speed of the ship is



(a) higher than it was before they started running.

(b) unchanged from what it was before they started running.

(c) lower than it was before they started running.

(d) impossible to determine.

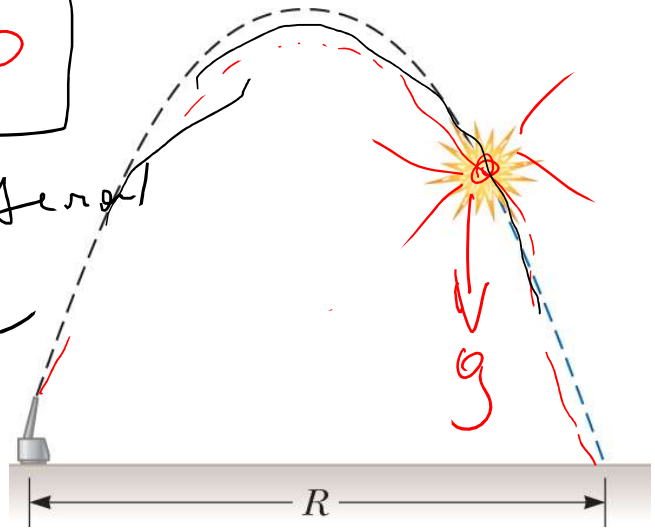
Conceptual Example 9.12: Exploding Projectile

A projectile fired into the air suddenly explodes into several fragments.

(**A**) What can be said about the motion of the center of mass of the system made up of all the fragments after the explosion?

$F_{ext} = \sum F_{internal} = 0$

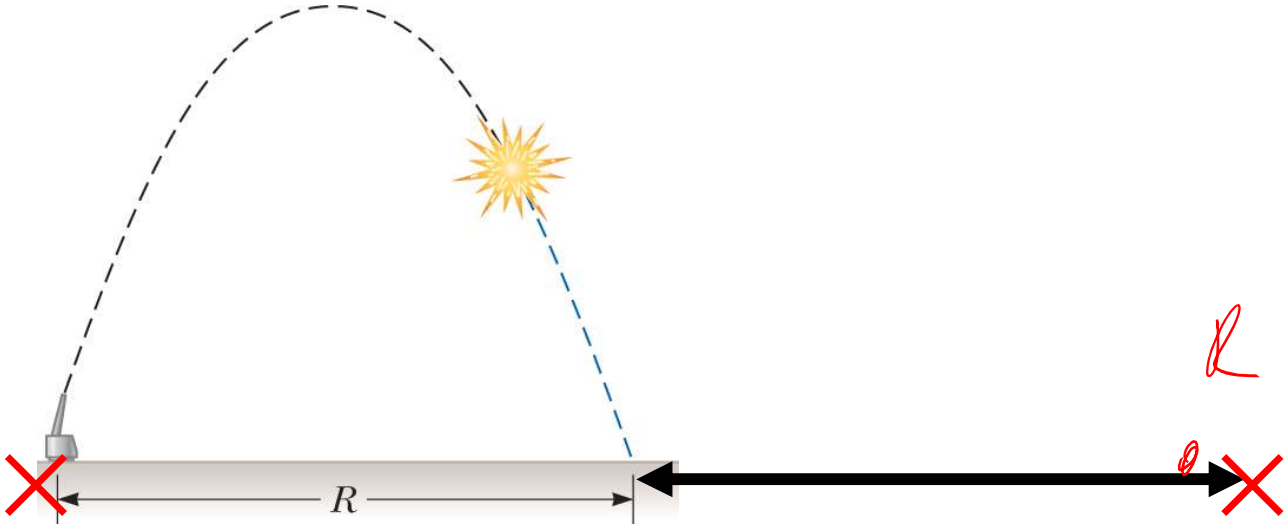
$F_{ext} \rightarrow$ gravitational force



Center of mass follows the same parabolic path as if no explosion had occurred.

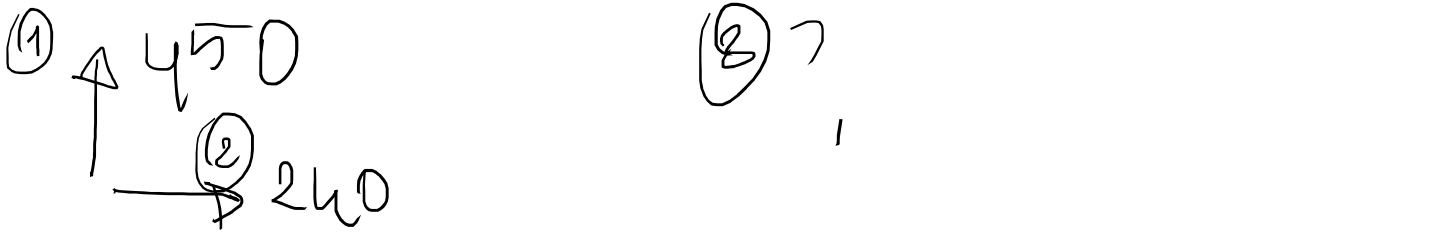
Conceptual Example 9.12: Exploding Projectile

(B) If the projectile did not explode, it would land at a distance R from its launch point. Suppose the projectile explodes and splits into two pieces of equal mass. One piece lands at a distance $2R$ to the right of the launch point. Where does the other piece land?



Example 9.13:
The Exploding Rocket

A rocket is fired vertically upward. At the instant it reaches an altitude of 1 000 m and a speed of v_i = 300 m/s, it explodes into three fragments having equal mass. One fragment moves upward with a speed of v_1 = 450 m/s following the explosion. The second fragment has a speed of v_2 = 240 m/s and is moving east right after the explosion. What is the velocity of the third fragment immediately after the explosion?



Example 9.13:
The Exploding Rocket

$$\Delta\vec{\mathbf{p}} = 0 \rightarrow \vec{\mathbf{p}}_i = \vec{\mathbf{p}}_f \rightarrow M\vec{\mathbf{v}}_i = \frac{M}{3}\vec{\mathbf{v}}_1 + \frac{M}{3}\vec{\mathbf{v}}_2 + \frac{M}{3}\vec{\mathbf{v}}_3$$

$$\vec{\mathbf{v}}_3 = 3\vec{\mathbf{v}}_i - \vec{\mathbf{v}}_1 - \vec{\mathbf{v}}_2$$

$$\vec{\mathbf{v}}_3 = 3\left(300\hat{\mathbf{j}}\ \text{m/s}\right) - \left(450\hat{\mathbf{j}}\ \text{m/s}\right) - \left(240\hat{\mathbf{i}}\ \text{m/s}\right)$$

$$= \left(-240\hat{\mathbf{i}} + 450\hat{\mathbf{j}}\right)\ \text{m/s}$$



Deformable Systems

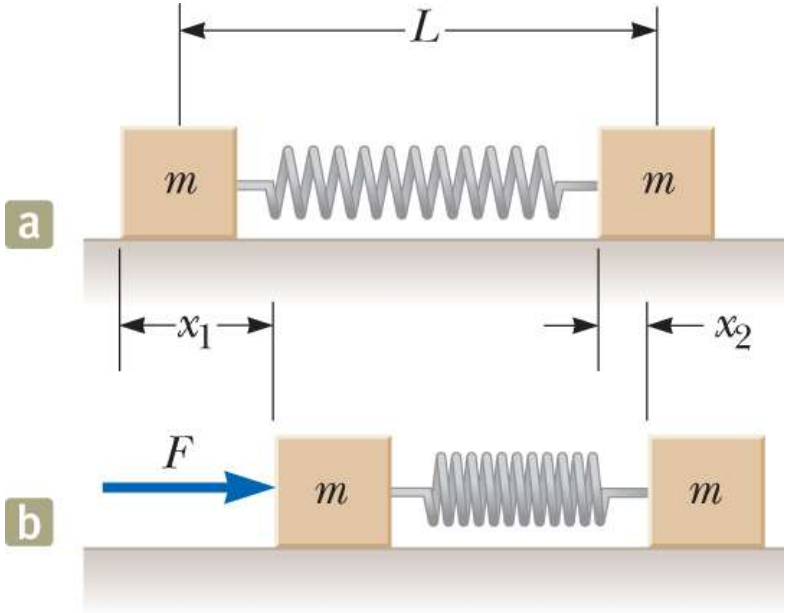
$$\Delta K + \Delta U = 0$$

$$\Delta \vec{\mathbf{p}}_{\text{tot}} = \vec{\mathbf{I}} \rightarrow m\Delta \vec{\mathbf{v}} = \int \vec{\mathbf{F}}_{\text{wall}} dt$$



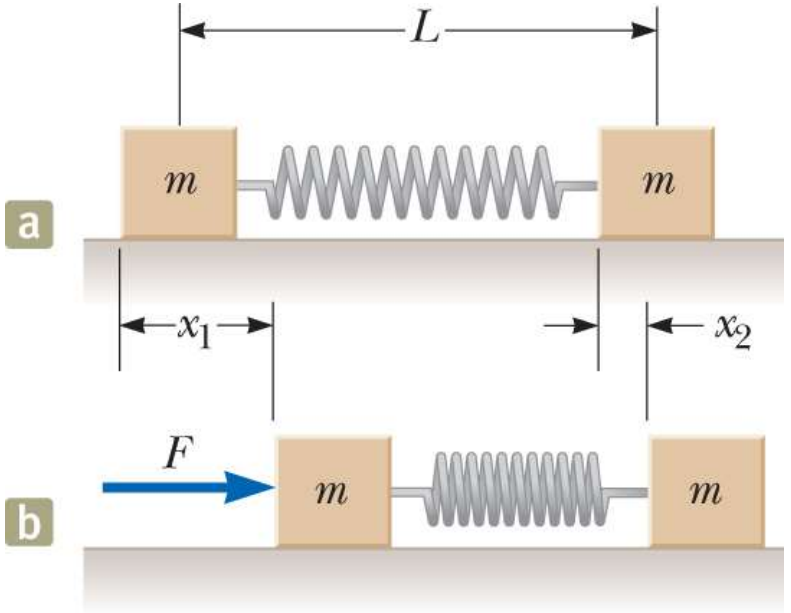
Example 9.14:
Pushing on a Spring

As shown in the top figure, two blocks are at rest on a frictionless, level table. Both blocks have the same mass m, and they are connected by a spring of negligible mass. The separation distance of the blocks when the spring is relaxed is L. During a time interval Δt, a constant force of magnitude F is applied horizontally to the left block, moving it through a distance x_1 as shown in the bottom figure. During this time interval, the right block moves through a distance x_2. At the end of this time interval, the force F is removed.



Example 9.14:
Pushing on a Spring

(**A**) Find the resulting speed $\mathbf{v}_{CM}$ of the center of mass of the system.



Example 9.15: Pushing on a Spring

$$\Delta p_x = I_x \rightarrow (2m)(v_{\text{CM}} - 0) = F\Delta t$$

$$2mv_{\text{CM}} = F\Delta t$$

$$\Delta t = \frac{\frac{1}{2}(x_1 + x_2)}{v_{\text{CM,avg}}}$$

$$\Delta t = \frac{\frac{1}{2}(x_1 + x_2)}{\frac{1}{2}(0 + v_{\text{CM}})} = \frac{(x_1 + x_2)}{v_{\text{CM}}}$$

$$2mv_{\text{CM}} = F\frac{(x_1 + x_2)}{v_{\text{CM}}} \rightarrow v_{\text{CM}} = \boxed{\sqrt{F\frac{(x_1 + x_2)}{2m}}}$$

Example 9.15: Pushing on a Spring

(**B**) Find the total energy of the system associated with vibration relative to its center of mass after the force F is removed.

$$K = K_{\text{CM}} + K_{\text{vib}}$$

$$\Delta K_{\text{CM}} + \Delta K_{\text{vib}} + \Delta U_{\text{vib}} = W$$

$$K_{\text{vib}} + U_{\text{vib}} = E_{\text{vib}} \rightarrow \Delta K_{\text{CM}} + \Delta E_{\text{vib}} = W$$

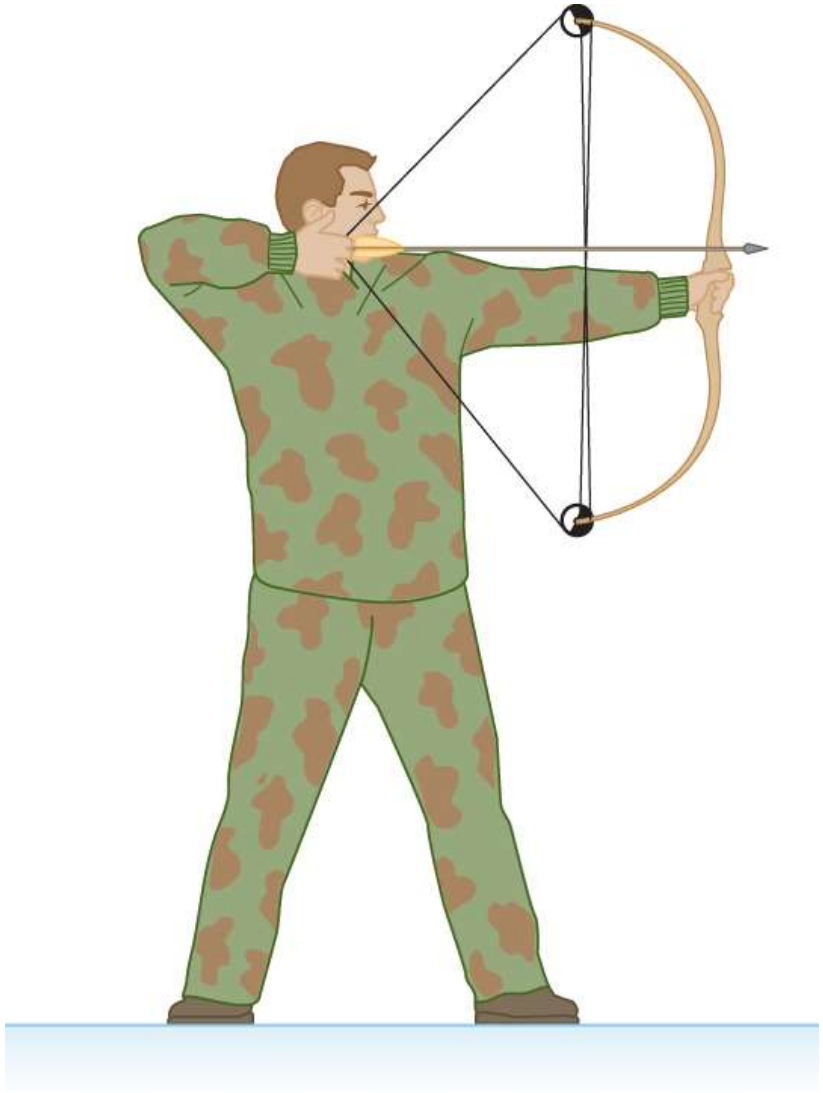
$$K_{\text{CM}} + E_{\text{vib}} = W = Fx_1$$

$$E_{\text{vib}} = Fx_1 - K_{\text{CM}} = Fx_1 - \frac{1}{2}(2m)v_{\text{CM}}{}^2 = \boxed{F\frac{(x_1 - x_2)}{2}}$$

Rocket Propulsion



Rocket Propulsion



Rocket Propulsion



a

Rocket Equation

$$\Delta p = 0 \rightarrow p_i = p_f \rightarrow Mv = \left(M - \Delta m\right)\left(v + \Delta v\right) + \Delta m\left(v - v_e\right)$$

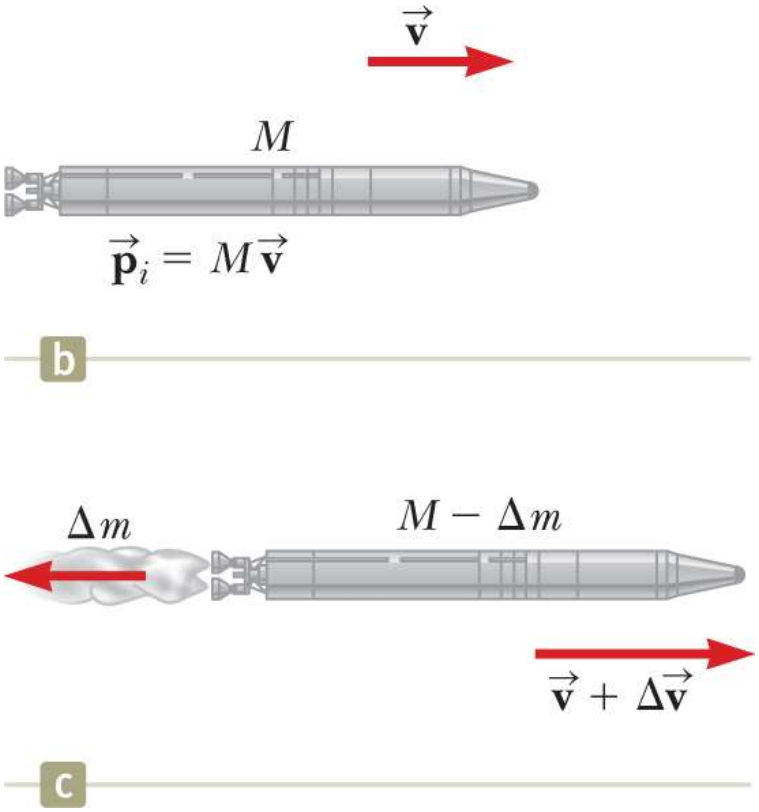
$$M\,\Delta v - \Delta m \Delta v = v_e \Delta m$$

$$\Delta v = \frac{v_e \Delta m}{M - \Delta m}$$

$$Mdv = v_e dm = -v_e dM$$

$$\int_{v_1}^{v_f} dv = -v_e \int_{M_i}^{M_f} \frac{dM}{M}$$

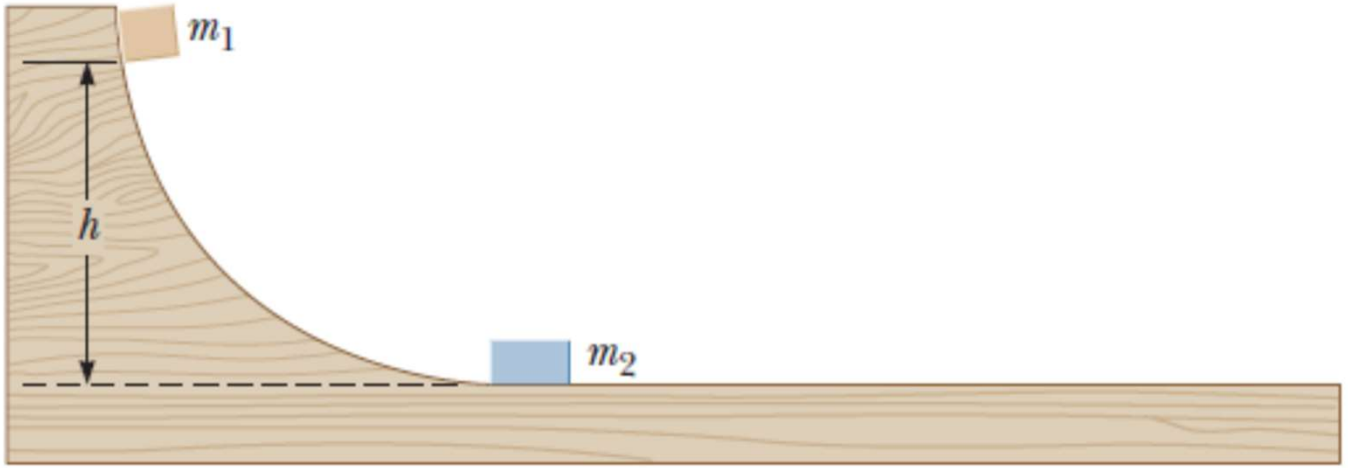
$$v_f - v_i = v_e \ln\left(\frac{M_i}{M_f}\right)$$



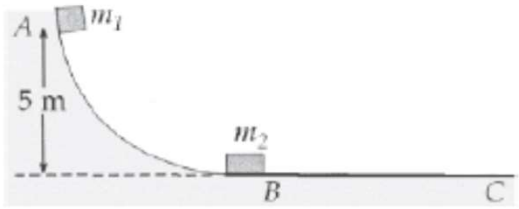
Thrust

$$\text{Thrust} = M\frac{dv}{dt} = \left| v_e \frac{dM}{dt} \right|$$

33. AMT W Two blocks are free to slide along the frictionless, wooden track shown in Figure P9.33. The block of mass $m_1 = 5.00$ kg is released from the position shown, at height $h = 5.00$ m above the flat part of the track. Protruding from its front end is the north pole of a strong magnet, which repels the north pole of an identical magnet embedded in the back end of the block of mass $m_2 = 10.0$ kg, initially at rest. The two blocks never touch. Calculate the maximum height to which m_1 rises after the elastic collision.



The mechanical energy of the isolated block-Earth system is conserved as the block of mass m_1 slides down the track. First we find v_1, the speed of m_1 at B before collision:



ANS. FIG. P9.33

$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}m_1 v_1^2 + 0 = 0 + m_1 g h$$

$$v_1 = \sqrt{2(9.80\ \text{m/s}^2)(5.00\ \text{m})} = 9.90\ \text{m/s}$$

Now we use the text's analysis of one-dimensional elastic collisions to find v_{1f}, the speed of m_1 at B just after collision.

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_1 = -\frac{1}{3}(9.90)\ \text{m/s} = -3.30\ \text{m/s}$$

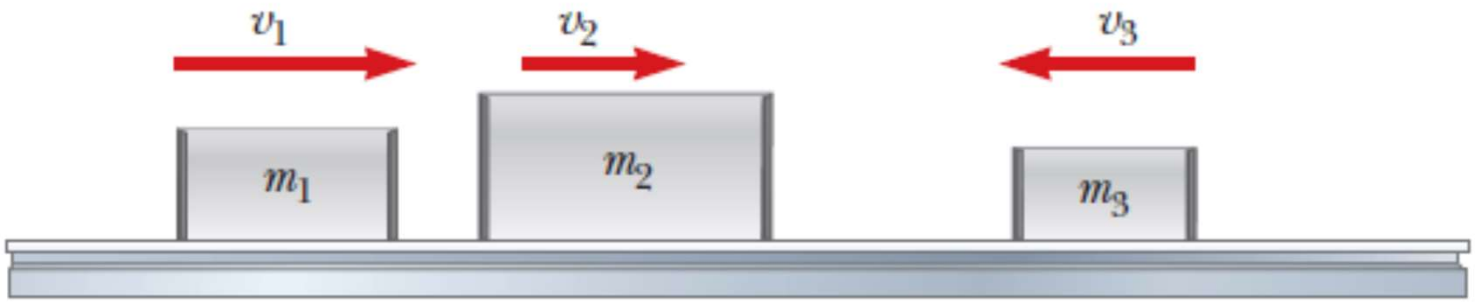
Now the 5-kg block bounces back up to its highest point after collision according to

$$m_1 g h_{\max} = \frac{1}{2} m_1 v_{1f}^2$$

which gives

$$h_{\max} = \frac{v_{1f}^2}{2g} = \frac{(-3.30\ \text{m/s})^2}{2(9.80\ \text{m/s}^2)} = \boxed{0.556\ \text{m}}$$

34. (a) Three carts of masses $m_1 = 4.00$ kg, $m_2 = 10.0$ kg, and $m_3 = 3.00$ kg move on a frictionless, horizontal track with speeds of $v_1 = 5.00$ m/s to the right, $v_2 = 3.00$ m/s to the right, and $v_3 = 4.00$ m/s to the left as shown in Figure P9.34. Velcro couplers make the carts stick together after colliding. Find the final velocity of the train of three carts. (b) **What If?** Does your answer in part (a) require that all the carts collide and stick



together at the same moment? What if they collide in a different order?

(a) Using conservation of momentum, $(\sum \vec{p})_{\text{before}} = (\sum \vec{p})_{\text{after}}$, gives

$$(4.00\ \text{kg})(5.00\ \text{m/s}) + (10.0\ \text{kg})(3.00\ \text{m/s}) + (3.00\ \text{kg})(-4.00\ \text{m/s}) = [(4.00 + 10.0 + 3.00)\ \text{kg}]v$$

Therefore, $v = +2.24$ m/s, or **2.24 m/s toward the right**.

(b) **No.** For example, if the 10.0-kg and 3.00-kg masses were to stick together first, they would move with a speed given by solving

$$(13.0\ \text{kg})v_1 = (10.0\ \text{kg})(3.00\ \text{m/s}) + (3.00\ \text{kg})(-4.00\ \text{m/s})$$

or $v_1 = +1.38$ m/s

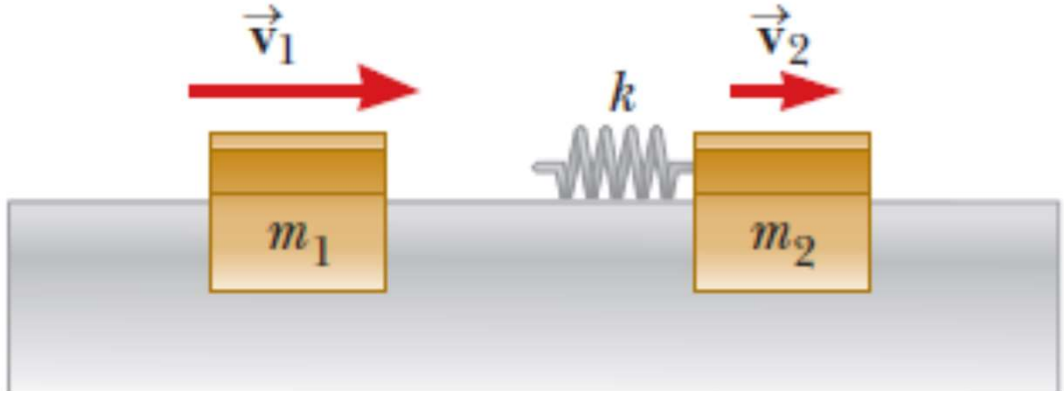
Then when this 13.0-kg combined mass collides with the 4.00-kg mass, we have

$$(17.0\ \text{kg})v = (13.0\ \text{kg})(1.38\ \text{m/s}) + (4.00\ \text{kg})(5.00\ \text{m/s})$$

and $v = +2.24$ m/s, just as in part (a).

Coupling order makes no difference to the final velocity.

75. Two gliders are set in motion on a horizontal air track. A spring of force constant k is attached to the back end of the second glider. As shown in Figure P9.75, the first glider, of mass m_1, moves to the right with speed v_1, and the second glider, of mass m_2, moves more slowly to the right with speed v_2. When m_1 collides with the spring attached to m_2, the spring compresses by a distance x_{max}, and the gliders then move apart again. In terms of v_1, v_2, m_1, m_2, and k, find (a) the speed v at maximum compression, (b) the maximum compression x_{max}, and (c) the velocity of each glider after m_1 has lost contact with the spring.



(a) When the spring is fully compressed, each cart moves with same velocity v. Apply conservation of momentum for the system of two gliders

$$p_i = p_f: \qquad m_1 v_1 + m_2 v_2 = (m_1 + m_2) v \quad \rightarrow \quad \boxed{v = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}}$$

(b) Only conservative forces act; therefore, $\Delta E = 0$.

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_1 v_2^2 = \frac{1}{2} (m_1 + m_2) v^2 + \frac{1}{2} k x_m^2$$

Substitute for v from (a) and solve for x_m.

$$x_m^2 = \left(\frac{1}{k(m_1 + m_2)} \right) [(m_1 + m_2) m_1 v_1^2 + (m_1 + m_2) m_2 v_2^2 - (m_1 v_1)^2 - (m_2 v_2)^2 - 2 m_1 m_2 v_1 v_2]$$

$$x_m = \sqrt{\frac{m_1 m_2 (v_1^2 + v_2^2 - 2 v_1 v_2)}{k(m_1 + m_2)}} = \boxed{(v_1 - v_2) \sqrt{\frac{m_1 m_2}{k(m_1 + m_2)}}}$$

(c) $m_1 v_1 + m_2 v_2 = m_1 v_{1f} + m_2 v_{2f}$

Conservation of momentum: $m_1\left(v_1 - v_{1f}\right) = m_2\left(v_{2f} - v_2\right)$ **[1]**

Conservation of energy: $\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_{2f}^2$

which simplifies to: $m_1\left(v_1^2 - v_{1f}^2\right) = m_2\left(v_{2f}^2 - v_2^2\right)$

Factoring gives

$$m_1\left(v_1 - v_{1f}\right)\left(v_1 + v_{1f}\right) = m_2\left(v_{2f} - v_2\right)\cdot\left(v_{2f} + v_2\right)$$

and with the use of the momentum equation (equation [1]),

this reduces to $v_1 + v_{1f} = v_{2f} + v_2$

or $v_{1f} = v_{2f} + v_2 - v_1$ **[2]**

Substituting equation [2] into equation [1] and simplifying yields

$$\boxed{v_{2f} = \frac{2m_1 v_1 + \left(m_2 - m_1\right)v_2}{m_1 + m_2}}$$

Upon substitution of this expression for into equation [2], one finds

$$\boxed{v_{1f} = \frac{\left(m_1 - m_2\right)v_1 + 2m_2 v_2}{m_1 + m_2}}$$